Two-colour crossing minimization in complete geometric graphs is NP-hard

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Abstract

Given a rational point set in general position, colour every edge of its complete straight-line graph with one of two colours and minimize the number of monochromatic properly crossing edge pairs. We prove that the decision problem is NP-complete, even for integer coordinates of polynomial binary length. A deterministic many-one reduction starts from MaxCut on represented permutation graphs. A double-chain comparison grid has two quantitatively stable phases; geometrically realized program points provide controlled fields on a source core. Convex separation excludes grid interactions with program nonfans, while the remaining auxiliary and completion costs are included explicitly. Writing $g(G) = |E(G)| - \mathrm{MaxCut}(G)$, the construction gives a computable baseline K_{base}, amplification Δ, and nonnegative errors satisfying

$$K_{\mathrm{base}} + \Delta g(G) - E_- \le \mathrm{OPT}(P) \le K_{\mathrm{base}} + \Delta g(G) + E_+,$$

with the strict gap

$$2^{18}(E_- + E_+) < \Delta.$$

The new analytic, geometric, and combinatorial parts of the reduction have been formalized in Lean. The formalization explicitly assumes the known convex complete-graph optimum formula; source NP-hardness and the machine-complexity argument are supplied separately. We state the verification boundary and the deterministic coordinate construction in detail.

Keywords. Complete geometric graph; monochromatic crossings; permutation graph; MaxCut; quantitative stability; Lean.

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1 Introduction

Let $P = (p_0, \dots, p_{N-1})$ be a labelled set of points in the plane. The complete geometric graph on P contains the straight segment $p_i p_j$ for every $i < j$. We assign one of two colours to every edge and count pairs of edges that cross properly and have the same colour. Both the point set and the drawing are fixed; the edge colours are the only variables.

The complexity of this problem for complete geometric graphs was posed at the Crossing Number Workshop in 2019 and appears as Vogtenhuber's Problem 2 in Section 4.12 of the report of Dagstuhl Seminar 21293 [3]. Completeness is a substantial restriction. A reduction that represents a hard graph by segments does not automatically survive adding every missing segment between their endpoints: the new edges are free variables and their optimal contribution can depend on the source colours. Appendix A gives a six-point example of this dependence. At the other extreme, for points in convex position the optimum is the known two-page crossing number of a complete graph [1].

Definition 1.1. The input to *complete geometric two-colour crossing minimization* is a list of pairwise distinct points $P \in (\mathbb{Q}^2)^N$ with no three collinear, together with an integer k in the decision version. Coordinates and k are encoded in binary. Write

$$E(P) = \{ij : 0 \le i < j < N\}, \qquad M(P, \chi) = \sum_{\{e,f\} \subseteq E(P)} \mathbf{1}[e \text{ and } f \text{ cross properly}]\, \mathbf{1}[\chi(e) = \chi(f)]$$

for $\chi : E(P) \to \{0, 1\}$, and put $\mathrm{OPT}(P) = \min_\chi M(P, \chi)$. The decision question is whether $\mathrm{OPT}(P) \le k$.

A proper crossing is an intersection of two open segments with four distinct endpoints. Several crossing pairs may have the same geometric intersection point; each pair is counted separately. The colouring is unrestricted: it is not a proper edge colouring in the graph-theoretic sense. There are no weights or prescribed colours.

Theorem 1.2. *The decision problem in Definition 1.1 is NP-complete under deterministic polynomial-time many-one reductions. Consequently, computing* $\mathrm{OPT}(P)$ *exactly is NP-hard. The decision problem remains NP-complete when all coordinates are integers encoded in binary.*

We reduce from MaxCut on permutation graphs, with a permutation representation supplied as part of the input. Its NP-completeness follows from de Figueiredo et al. [2]: their construction explicitly produces the two permutations. After relabelling the first order, a source graph can be described by a permutation π of $[n]$, with an edge ij when $i < j$ and $\pi(i) > \pi(j)$. For a source graph G, set

$$g(G) = \min_{x : V(G) \to \{0,1\}} \sum_{uv \in E(G)} \mathbf{1}[x(u) = x(v)] = |E(G)| - \mathrm{MaxCut}(G).$$

Thus a source threshold k becomes the equivalent integer threshold $g(G) \le t$, where $t = |E(G)| - k$.

Our construction has three interacting components. A large rectangular comparison grid forces one of two stable colour phases. Small families of geometric program points then produce controlled fields on the edges encoding the source. Finally, an exact convex separation removes the potentially dominant interaction between the grid and program nonfans. The remaining added edges are included in an explicit error bound. The conclusion is a uniform interval

$$K_{\mathrm{base}} + \Delta g(G) - E_- \;\le\; \mathrm{OPT}(P) \;\le\; K_{\mathrm{base}} + \Delta g(G) + E_+, \qquad 2^{18}(E_- + E_+) < \Delta, \tag{1.1}$$

where K_{base} is computable without an optimization oracle. The lower bound holds for an unrestricted optimum of the complete graph; the upper bound is attained within the stated error by a single complete colouring. No colour restrictions are part of the output instance.

1.1 Conventions and scales

We use N for the number of points of the final geometric instance. The letter N_g will denote the number of vertices of a comparison grid when a distinction is needed. For nonnegative integers v, define

$$Z(v) = \frac{1}{4}\left\lfloor \frac{v}{2} \right\rfloor \left\lfloor \frac{v-1}{2} \right\rfloor \left\lfloor \frac{v-2}{2} \right\rfloor \left\lfloor \frac{v-3}{2} \right\rfloor \quad (v \geq 3), \qquad Z(0) = Z(1) = Z(2) = 0.$$

The theorem of Ábrego et al. [1, Theorem 7] says that $Z(v)$ is the optimum for a convex complete geometric graph: a same-page pair of edges in a two-page drawing crosses exactly when its endpoints alternate in the linear order. Boundary edges do not affect this count. We use this established value for two convex background components.

For a source with $n \geq 1$, the parameters are

$$\begin{gathered} q = 2^{20}(n+1), \quad S = nq, \quad d = S+1, \quad H = 2^{20000}d^{40}, \\ W = H^4, \quad B = H^5, \quad D_m = 512SW, \quad o = 2SW + D_m = 514SW, \\ T = 2^{13}o^2, \qquad A_0 = 2^{13}dT, \qquad \Delta = q^2W^4. \end{gathered} \tag{1.2}$$

All exponents and constants are absolute. These deliberately generous scales simplify exact error separation; no practical implementation of the full, extremely large target instances is claimed.

1.2 Membership in NP and integer coordinates

Proposition 1.3. *The problem in Definition 1.1 belongs to NP. Rational and integer coordinate versions are polynomial-time equivalent.*

Proof. Let L be the total input bit length. Normalize denominators to be positive and let D be the product of all coordinate denominators. Its bit length is $O(L)$. Multiplying every coordinate by this same positive integer yields integer coordinates of $O(L)$ bits and preserves all orientation signs. Computing D and the scaled coordinates takes polynomial time by ordinary integer arithmetic.

Pairwise distinctness and absence of collinear triples can be checked by examining point pairs and the determinants

$$\mathrm{or}(a,b,c) = (b_x - a_x)(c_y - a_y) - (b_y - a_y)(c_x - a_x).$$

Declare invalid inputs to be no-instances. A certificate consists of the $\binom{N}{2}$ edge colours. For every unordered pair ab, cd with four distinct endpoints, a proper crossing is equivalent to

$$\mathrm{or}(a,b,c)\,\mathrm{or}(a,b,d) < 0, \qquad \mathrm{or}(c,d,a)\,\mathrm{or}(c,d,b) < 0.$$

There are $O(N^4)$ such tests, each using a constant number of operations on integers of polynomial bit length. Count those crossing pairs with equal colours and compare the result with k. Negative thresholds are immediate no-instances. This verifies a certificate in polynomial time. The same positive scaling is a reduction from rational to integer inputs; the converse inclusion is immediate. □

The rest of the paper establishes the reduction and the quantitative interval (1.1). The formalization and its precise external dependencies are described in Section 6.

2 Stability of the comparison grid

The comparison grid supplies a reference colouring without prescribing any target edge colours. We first prove a quantitative continuous statement, then transfer it to every finite rectangular grid. The finite statement applies to actual minimizers with arbitrary bounded unary perturbations. A separate estimate applies to any colouring with the resulting localized support; this distinction will allow us to change the program colours while keeping the same grid colouring.

2.1 The continuous comparison energy

All integrals in this section use Lebesgue measure. Functions are Borel measurable; the statements also hold for Lebesgue-measurable functions after replacement on a null set. For a sign function $f : [0,1]^2 \to \{-1,1\}$, put

$$\mathcal{E}(f) = \int_{\substack{0\le u<u'\le 1\\ 0\le v<v'\le 1}} \mathbf{1}\{f(u,v) = f(u',v')\}\, du\, dv\, du'\, dv'. \tag{2.1}$$

Each comparable unordered pair occurs once. Define

$$f_*(u,v) = \begin{cases} 1, & u+v<1,\\ -1, & u+v\ge 1. \end{cases}$$

The diagonal convention has no effect on the continuous integral, but we retain this strict inequality when passing to finite grids.

Theorem 2.1 (Continuous two-phase stability)**.** *Every measurable sign function on the unit square has $\mathcal{E}(f) \ge 1/12$. For every $\varepsilon \ge 0$, if $\mathcal{E}(f) \le 1/12 + \varepsilon$, then*

$$\min_{\sigma\in\{-1,1\}} \|f - \sigma f_*\|_{L^1([0,1]^2)} \le 4096\varepsilon^{1/72}. \tag{2.2}$$

We give the analytic ingredients with constants, since only a uniform estimate, rather than a sharp exponent, is needed later.

For $h : [0,1] \to \{-1,1\}$ define

$$(\mathcal{T}h)(t) = \int_0^1 \operatorname{sgn}(s-t)h(s)\, ds, \qquad D(h) = \frac{1}{3} - \|\mathcal{T}h\|_2^2.$$

Values of the sign kernel on its diagonal are immaterial. A threshold function is a sign function that is constant on each side of a point $c \in [0,1]$, with opposite signs on those two intervals. Both orientations and, through $c = 0, 1$, both constant functions are included.

Lemma 2.2 (One-dimensional stability)**.** *For every measurable sign function h, one has $D(h) \ge 0$. There is a threshold function b such that*

$$\|h - b\|_1 \le 4D(h)^{1/3}.$$

Proof. Let

$$I(h) = \int_{0<r<s<t<1} \mathbf{1}\{h(r) = h(t) \ne h(s)\}\, dr\, ds\, dt.$$

The elementary identity

$$4\mathbf{1}\{h(r) = h(t) \ne h(s)\} = 1 + h(r)h(t) - h(r)h(s) - h(s)h(t)$$

gives

$$D(h) = 8I(h). \tag{2.3}$$

To check the coefficient, after integrating out the third variable the coefficient of $h(u)h(v)$, for $u < v$, in the triple expression is $2(v-u)-1$. On the other hand,

$$\int_0^1 \operatorname{sgn}(u-t)\operatorname{sgn}(v-t)\,dt = 1 - 2|u-v|.$$

The constant triple integral is $1/6$, and the ordered two-variable integral is half the corresponding symmetric integral. These observations give (2.3), including its constant term, by Fubini. All integrands are bounded, so these changes of order are legitimate.

Let e be the least measure of disagreement with a threshold. For each orientation, disagreement is continuous in its threshold position, and $[0,1]$ is compact, so the minimum is attained. We claim $I(h) \geq e^3/64$. There is nothing to prove if $e = 0$. Otherwise let p be the measure of the positive set. The constant negative threshold shows that $p \geq e$. By continuity of cumulative positive mass choose $a \leq b$ so that the positive mass before a and after b is $e/4$ each.

If the negative mass between a and b is at least $e/4$, independently choosing a positive point before a, a negative point between a, b, and a positive point after b gives $I(h) \geq (e/4)^3$. Otherwise compare with the threshold positive before b and negative after b. Its disagreement is at least e, whereas the negative mass between a, b plus the positive mass after b is less than $e/2$. Thus the negative mass before a is at least $e/2$. Comparing with the oppositely oriented threshold at a similarly gives negative mass at least $e/2$ after b. The positive mass between a, b is $p - e/2 \geq e/2$. Negative–positive–negative triples therefore give the stronger bound $I(h) \geq e^3/8$. The claim follows. By (2.3), $e^3 \leq 8D(h)$; since the sign-function L^1 distance is $2e$, the asserted bound follows. □

Apply $\mathcal{T}$ in the two coordinates and write

$$A(u,v) = \int_0^1 \operatorname{sgn}(s-u) f(s,v)\,ds, \qquad C(u,v) = \int_0^1 \operatorname{sgn}(s-v) f(u,s)\,ds, \qquad M = \int_{[0,1]^2} f.$$

Expanding the order indicators and the equality indicator, and using antisymmetry of the sign kernel, gives

$$\mathcal{E}(f) = \frac{1 + M^2 - \langle A, C\rangle}{8}.$$

In particular,

$$8(\mathcal{E}(f) - 1/12) = M^2 + \frac{1}{2}\|A - C\|_2^2 + \frac{1}{2}\left(\frac{1}{3} - \|A\|_2^2\right) + \frac{1}{2}\left(\frac{1}{3} - \|C\|_2^2\right). \tag{2.4}$$

Lemma 2.2 on each row and column makes every term on the right nonnegative. This proves the lower bound in Theorem 2.1. For f_* the mean vanishes, both slice families are thresholds, and direct integration gives $A = C$. Hence its energy is exactly $1/12$.

Under an excess bound ε, (2.4) gives

$$|M| \leq \sqrt{8\varepsilon}, \qquad \|A - C\|_2 \leq 4\sqrt{\varepsilon}, \qquad \int_0^1 D(f(\cdot, v))\,dv \leq 16\varepsilon, \tag{2.5}$$

and the analogous column estimate. Jensen's inequality and Lemma 2.2 show that the average row distance from a threshold, and also the average column distance, is at most $4(16\varepsilon)^{1/3} \leq 11\varepsilon^{1/3}$. These distance functions are measurable: one may take the infimum over rational threshold positions in both orientations, using continuity in the threshold position. No measurable selection of optimal thresholds is required.

Lemma 2.3 (Approximation by a diagonal profile)**.** *Suppose* $0 < h \le 1$ *and* $\mathcal{E}(f) \le 1/12 + h^6$. *There is a measurable* $b : [0,2] \to \{-1,1\}$ *such that, for* $f_1(u,v) = b(u+v)$,

$$\|f - f_1\|_1 \le 128h, \qquad \mathcal{E}(f_1) \le 1/12 + 257h.$$

Proof. It suffices first to treat $h < 1/8$. Extend f by zero outside the square. A threshold on $[0,1]$, extended by zero, has total variation at most four. Approximating each row by a threshold and applying the triangle inequality shows that translation by r in the first coordinate changes the zero extension in L^1 by at most $22\varepsilon^{1/3} + 4|r|$. The same holds in the second coordinate. Here the rowwise inequality can be integrated using the measurable infimum just described, so no selection issue arises.

Let f_h be the convolution of the zero extension with the uniform probability density on $[-h,h]^2$. Integrating the translation bound and using $\varepsilon = h^6$ yields

$$\|f - f_h\|_{L^1(\mathbb{R}^2)} \le 44h^2 + 8h \le 52h. \tag{2.6}$$

On the interior square $Q_h = [h, 1-h]^2$ we also have

$$\|(\partial_u - \partial_v) f_h\|_{L^1(Q_h)} \le 2h. \tag{2.7}$$

Indeed, on the open unit square, in the sense of distributions,

$$(\partial_u - \partial_v) f = \tfrac{1}{2}\partial_u \partial_v (A - C),$$

because $\partial_u A = -2f$ and $\partial_v C = -2f$. The mixed derivative of the uniform averaging kernel is the signed measure at its four corners with total variation h^{-2}. Convolution on Q_h therefore gives a bound $\|A - C\|_1/(2h^2) \le 2h$ by (2.5). Only averaging boxes contained in the original square are used here, so no distributional boundary term from the zero extension enters this estimate. The box average of a bounded function is Lipschitz: translating either coordinate changes the averaging rectangle by area at most a constant times the translation distance. Consequently the weak directional derivative above agrees with its almost-everywhere derivative, and the one-dimensional fundamental theorem applies on almost every diagonal slice.

For each s average $f_h(u, s-u)$ over the interval of u for which $(u, s-u) \in Q_h$, and denote the average by $j(s)$. Put $j(s) = 0$ when that interval is empty. This defines a measurable function with $|j| \le 1$. On an interval I of length ℓ, an absolutely continuous function g satisfies

$$\int_I |g - g_I| \le \frac{1}{\ell} \int_I \int_I |g(x) - g(y)|\, dx\, dy \le \ell \int_I |g'|.$$

The diagonal coordinates (s,u) have Jacobian one and their intervals have length at most one. Integrating this inequality and using (2.7) gives $\|f_h - j(u+v)\|_{L^1(Q_h)} \le 2h$. In particular no inverse lower bound on the length of a short diagonal is required. Outside Q_h the area is at most $4h$ and the pointwise difference is at most two. Together with (2.6), this gives

$$\|f - j(u+v)\|_{L^1([0,1]^2)} \le 52h + 2h + 8h = 62h.$$

Round j to a sign, with a fixed convention at zero. For a sign z and $r \in [-1,1]$, $|z - \operatorname{sgn}(r)| \le 2|z - r|$. Thus the resulting f_1 has distance at most $124h \le 128h$.

For any two sign functions f, g, expansion of the equality indicator gives the useful bound

$$|\mathcal{E}(f) - \mathcal{E}(g)| \le \frac{1}{2} \int_{[0,1]^2} |f - g|(uv + (1-u)(1-v))\, du\, dv \le \|f - g\|_1. \tag{2.8}$$

Hence $\mathcal{E}(f_1) \le 1/12 + h^6 + 128h \le 1/12 + 129h$, which is stronger than the stated bound. If $h \ge 1/8$, any fixed diagonal sign function suffices: the distance is at most two, the energy is at most $1/4$, and both claimed bounds follow. This completes the lemma for $0 < h \le 1$. □

Proof of Theorem 2.1. Assume first that $\varepsilon > 0$, put $h = \varepsilon^{1/6}$, and suppose $\alpha = 257h \le 1/4096$. Lemma 2.3 gives $f_1(u,v) = b(u+v)$ with energy excess at most α. Let $t = \alpha^{1/4} \le 1/8$. By (2.5) there is a row at height $v \in [t, 2t]$ whose one-dimensional deficit is at most $16\alpha/t$. Its sign profile is b restricted to $[v, v+1]$. By Lemma 2.2, it has a threshold approximation of error at most $4(16\alpha/t)^{1/3} < 11\alpha^{1/4}$. Restrict that threshold to $[v, 1]$ and extend it to $[0, 1]$ as a threshold. The additional interval has length at most $2t$ and costs at most $4t$. Thus b on $[0, 1]$ is within $15\alpha^{1/4}$ of a threshold b_0. Selecting a row with $v \in [1-2t, 1-t]$ in the same way gives a threshold b_1 within $15\alpha^{1/4}$ of $s \mapsto b(1+s)$ on $[0, 1]$.

Let $\widetilde{b}(s) = b_0(s)$ for $s < 1$ and $\widetilde{b}(s) = b_1(s-1)$ for $s \ge 1$, and put $f_2(u,v) = \widetilde{b}(u+v)$. The density of $u+v$ on the square is s for $0 \le s \le 1$ and $2-s$ for $1 \le s \le 2$, so it is at most one. Consequently

$$\|f_1 - f_2\|_1 \le \zeta := 30\alpha^{1/4}. \tag{2.9}$$

Each coordinate operator $\mathcal{T}$ has L^2 operator norm at most one, by Cauchy–Schwarz on each unit interval and Fubini. Moreover, $\|f_1 - f_2\|_2^2 = 2\|f_1 - f_2\|_1$. Writing $\mathcal{D} = \mathcal{T}_u - \mathcal{T}_v$, the triangle inequality followed by $(x+y)^2 \le 2x^2 + 2y^2$ gives

$$\|\mathcal{D} f_2\|_2^2 \le 32\alpha + 16\zeta \le 512\alpha^{1/4}. \tag{2.10}$$

Set $m_0 = |\{s \in [0,1] : b_0(s) = b_1(s)\}|$ and $F(t) = \int_0^t (b_0(s) + b_1(s))\, ds$. Directly integrating the two coordinate operators yields

$$\mathcal{D} f_2(u,v) = F(v) - F(u), \qquad \|\mathcal{D} f_2\|_2^2 = 2\int_0^1 (F - \overline{F})^2. \tag{2.11}$$

Because b_0, b_1 are thresholds, the nonzero derivative of F is supported on at most two intervals, and is constantly 2 or -2 on each. If $m_0 > 0$, one of those intervals has length $\ell \ge m_0/2$. For any constant c, the integral of $(F-c)^2$ on that interval is at least

$$\min_d \int_0^\ell (2s - d)^2\, ds = \ell^3/3.$$

Taking $c = \overline{F}$ in (2.11) and using (2.10), we obtain $m_0^3 \le 6144\alpha^{1/4}$ and hence

$$m_0 \le 19\alpha^{1/12}. \tag{2.12}$$

Replace b_1 by $-b_0$ and call the resulting diagonal function f_3. Since the diagonal density is at most one, $\|f_2 - f_3\|_1 \le 2m_0$. If b_0 is positive before c and negative after c, then

$$\int f_3 = \int_0^1 (2s-1) b_0(s)\, ds = -2c(1-c).$$

Reversing the threshold orientation reverses the mean. The distances from f_3 to the two canonical phases are $2c$ and $2(1-c)$, in some order, since the density weights of the two halves add to one. Therefore its distance to a canonical phase is at most $2|\int f_3|$. With $d = \|f - f_2\|_1$, the triangle inequality now gives

$$\min_\sigma \|f - \sigma f_*\|_1 \le 3d + 6m_0 + 2\left|\int f\right|.$$

Using $d \le 128h + 30\alpha^{1/4}$, (2.12), and $|\int f| \le 3h^3$, the right-hand side is at most

$$384h + 90\alpha^{1/4} + 114\alpha^{1/12} + 6h^3 \le 594\alpha^{1/12}.$$

Here $0 < h \le 1$, $\alpha = 257h \le 1$, and each smaller power term is bounded by the displayed coefficient times $\alpha^{1/12}$. Since $257^{1/12} \le 2$, this is at most $1188h^{1/12} \le 4096\varepsilon^{1/72}$.

If $257h > 1/4096$, the distance between any two sign functions is at most two. The inequality $h^{1/12} \ge 1/4$ at the cutoff, and monotonicity thereafter, give the asserted estimate immediately. Finally, if $\varepsilon = 0$, apply the positive-excess result for every positive excess parameter. The minimum of the distances to the two fixed phases must then be zero. One of those two phases attains that minimum, proving the zero-excess case without a choice of a limiting phase. □

2.2 Finite rectangular grids with unary costs

For integers $m, n \ge 1$, let $J_{m,n}$ have vertex set $\{0, \dots, m-1\} \times \{0, \dots, n-1\}$, where two vertices are adjacent exactly when both coordinate orders strictly agree. In this section write

$$V = mn, \qquad \rho = \frac{1}{m} + \frac{1}{n}, \qquad u_i = \frac{i + 1/2}{m}, \qquad v_j = \frac{j + 1/2}{n}.$$

Thus V is the number of vertices of the comparison graph, not the number of points of the eventual geometric drawing. Define

$$c^+_{ij} = \mathbf{1}\{u_i + v_j < 1\}, \qquad c^-_{ij} = 1 - c^+_{ij}, \qquad s^\sigma_{ij} = 2c^\sigma_{ij} - 1.$$

In particular a diagonal cell centre has spin -1 in the positive phase.

Let $U_x : \{0, 1\} \to \mathbb{R}$ be unary costs, and suppose

$$|U_x(1) - U_x(0)| \le \alpha V \quad \text{for all } x, \qquad \alpha \ge 0.$$

No bound is imposed on their additive constants. Write

$$E(c) = \Phi_J(c) + \sum_x U_x(c_x),$$

where Φ_J counts monochromatic edges of $J_{m,n}$. For a set F of grid vertices, $c \oplus F$ denotes the colouring obtained by flipping exactly F. Define the exact finite quantities

$$d^\sigma_x = E(c^\sigma \oplus \{x\}) - E(c^\sigma), \qquad L_\sigma = E(c^\sigma) + \sum_x \min(0, d^\sigma_x), \qquad L = \min(L_+, L_-). \tag{2.13}$$

For $w \ge 0$ let

$$D_w = \{(i, j) : |u_i + v_j - 1| \le w\}, \qquad C_w = \{(i, j) : u_i + v_j \le w \text{ or } u_i + v_j \ge 2 - w\}.$$

Theorem 2.4 (Finite-grid localization and energy bounds)**.** *Let a satisfy $\alpha + \rho \le a \le 2^{-4096}$. For every rectangular aspect ratio the following statements hold.*

1. *Every minimizer c of E has a phase σ such that, with $F = \{x : c_x \ne c^\sigma_x\}$,*

$$|F| \le 52aV, \qquad F \subseteq D_{220a} \cup C_{220a}.$$

2. *The minimum energy belongs to*

$$[L - 2^{32}a^3V^2,\ L + 2^{32}a^3V^2].$$

For either prescribed phase, the explicit set $F^0_\sigma = \{x : d^\sigma_x < 0\}$ has size at most $52aV$ and

$$E(c^\sigma \oplus F^0_\sigma) \le L_\sigma + 2^{32}a^3V^2.$$

3. *For every* $F \subseteq D_{220a} \cup C_{220a}$,

$$E(c^\sigma \oplus F) \geq L_\sigma - 2^{32}a^3V^2.$$

This last conclusion requires no optimality of $c^\sigma \oplus F$. *In fact it holds for arbitrary real unary costs, without their oscillation bound, whenever* $\rho \leq a \leq 1/65536$.

For rational unary costs, both L_σ *and the two prescribed-phase colourings are computable by polynomially many exact arithmetic operations.*

Proof. From cells to the continuous objective. Extend a grid colouring to the sign function f_c constant on its $1/m$ by $1/n$ cells. Comparable pairs in distinct rows and columns contribute exactly $\Phi_J(c)/V^2$. All other contributions are nonnegative and their total uncoloured measure is

$$\frac{1}{4} - \frac{\binom{m}{2}\binom{n}{2}}{m^2n^2} = \frac{1}{4m} + \frac{1}{4n} - \frac{1}{4mn} \leq \frac{\rho}{4}.$$

We will use the weaker bound

$$0 \leq \mathcal{E}(f_c) - \Phi_J(c)/V^2 \leq \rho/2. \tag{2.14}$$

The functions f_{c^+} and f_* can disagree only where $|u + v - 1| \leq \rho/2$. This band has area at most ρ, whence $\|f_{c^+} - f_*\|_1 \leq 2\rho$, and the analogous assertion holds for the negative phase. By (2.8), $\mathcal{E}(f_{c^+}) \leq 1/12 + 2\rho$. For a minimizer c, comparison with c^+ gives $\Phi_J(c) \leq \Phi_J(c^+) + \alpha V^2$. Therefore

$$\mathcal{E}(f_c) \leq 1/12 + \alpha + \tfrac{5}{2}\rho \leq 1/12 + 5a.$$

Theorem 2.1 and the canonical cell discrepancy give a phase whose disagreement fraction $\mu = |F|/V$ satisfies

$$\mu \leq 2048(5a)^{1/72} + \rho \leq 2^{-20}. \tag{2.15}$$

For the final numerical bound, use $5 < 8$, $a \leq 2^{-4096}$, and $\rho \leq a$; the first term is less than $2^{11-4093/72} < 2^{-45}$. This initial estimate follows from the actual minimum condition.

Canonical local fields. For $p = (u, v)$ let Q_p be the union of the lower-left and upper-right rectangles comparable to p. With $s = 1 - u - v$, direct integration gives

$$\int_{Q_p} f_* = -s(1 - |s|).$$

For $s \geq 0$, the lower-left rectangle contributes uv, and the upper-right rectangle contributes $s^2 - (1-u)(1-v)$; reflection gives the other case. The union of cells corresponding to the finite neighbours of a cell centre differs from Q_p only in its row and column strips, of total area at most ρ. The sign-function discrepancy contributes at most another 2ρ. A single grid flip changes the cost by minus the vertex spin times its neighbour-spin sum. Consequently, in either canonical phase,

$$\left| \frac{d_x^{\mathrm{grid}}}{V} - \ell(u_i, v_j) \right| \leq 6\rho, \qquad \ell(u, v) = |1 - u - v|(1 - |1 - u - v|). \tag{2.16}$$

At a diagonal centre the continuous field is zero, so the strict tie convention causes no change to this bound. Including the unary cost gives

$$d_x^\sigma \geq V\ell(u_i, v_j) - 6aV. \tag{2.17}$$

If $t \in [0,1]$ and $t(1-t) \le z$, then either $t \le 2z$ or $1-t \le 2z$, by considering $t \le 1/2$ and $t \ge 1/2$ separately. Thus a small value of ℓ places the vertex in a narrow diagonal strip or in the two corner regions.

Finite strip and corner counts. A rectangle with side lengths r, t contains at most

$$(mr+1)(nt+1) \le V(r+\rho)(t+\rho)$$

cell centres. For $\rho \le w \le 1$, counting by rows gives $|D_w| \le (2w+\rho)V \le 3wV$, and bounding each corner triangle by a square gives $|C_w| \le 8w^2V$. Two comparable points of D_w, ordered coordinatewise, have the sum of their coordinate differences at most $2w$. For any strip point its strip neighbours therefore lie in two squares of side $2w$, giving at most $18w^2V$ possibilities. Hence for every $F \subseteq D_w \cup C_w$,

$$\deg_F(x) \le 26w^2V \quad (x \in D_w), \qquad |E(J_{m,n}[F])| \le 83w^3V^2. \tag{2.18}$$

For the edge bound, strip–strip, strip–corner, and corner–corner edges contribute at most $27w^3V^2$, $24w^3V^2$, and $32w^4V^2$, respectively. Counting an edge more than once when regions overlap only increases these upper bounds. The degree assertion is made only for strip vertices: corner vertices need not have degree of order w^2V.

Bootstrap from initial stability. For a disagreeing vertex $x \in F$, changing its colour back in the minimizer has cost change

$$-d_x^\sigma - 2 \sum_{y \in F:\, xy \in E(J_{m,n})} s_x^\sigma s_y^\sigma \ge 0.$$

Thus $d_x^\sigma \le 2\deg_F(x)$, and (2.17) yields

$$\ell(u_i, v_j) \le 6a + 2\deg_F(x)/V \qquad (x \in F). \tag{2.19}$$

Initially $\deg_F(x) \le \mu V$. The observation following (2.17) gives

$$F \subseteq D_w \cup C_w, \qquad w = 4\mu + 12a.$$

By (2.15), $\rho \le w < 1/4$. For $x \in F \cap D_w$, (2.18) improves (2.19) to $\ell \le 6a + 52w^2$. In D_w we have $\ell \ge |1-u-v|/2$, so $F \cap D_w \subseteq D_{12a+104w^2}$. The direct row count and the corner count now imply

$$\mu \le 25a + 216w^2 \le 25a + 6912\mu^2 + 62208a^2.$$

Since $\mu \le 2^{-20}$ and $a \le 2^{-4096}$, we have $6912\mu^2 \le \mu/2$ and $62208a^2 \le a$. Absorbing these terms proves $\mu \le 52a$. Substitution in the definition of w gives $w \le 220a$, establishing the first conclusion. No small degree bound for corner vertices has been used in this absorption.

Exact correction for simultaneous flips. For every flip set F and arbitrary real unary costs,

$$E(c^\sigma \oplus F) - E(c^\sigma) = \sum_{x \in F} d_x^\sigma + 2 \sum_{\{x,y\} \in E(J_{m,n}[F])} s_x^\sigma s_y^\sigma. \tag{2.20}$$

Indeed, an edge with both endpoints flipped keeps its equality status, while its two single-flip contributions sum to $-2s_x^\sigma s_y^\sigma$. The second sum restores exactly this discrepancy; unary terms already add independently.

If $F \subseteq D_{220a} \cup C_{220a}$ and $\rho \le a \le 1/65536$, (2.18) bounds the absolute correction by

$$2 \cdot 83 \cdot 220^3 a^3 V^2 < 2^{32} a^3 V^2.$$

The linear sum is at least $\sum_x \min(0, d_x^\sigma)$. This proves the third conclusion, including its stronger independence from the unary oscillation and from optimality. Applying it to the localized minimizer proves the lower bound in the second conclusion.

For the upper bound in a prescribed phase, take exactly $F_\sigma^0 = \{x : d_x^\sigma < 0\}$. By (2.17), this set is contained in $D_{12a} \cup C_{12a}$, and its cardinality is at most

$$(36a + 1152a^2)V \le 52aV.$$

Its linear sum in (2.20) is the defining sum for L_σ, while its correction is bounded by $2 \cdot 83 \cdot 12^3 a^3 V^2 < 2^{32} a^3 V^2$. Choosing the phase with smaller L_σ proves the upper bound for the minimum. Finally, all quantities in (2.13) are finite sums and single-vertex comparisons. They require no search over colourings and can be evaluated exactly in polynomially many operations for rational unary costs. □

2.3 Freezing probes while preserving the actual grid

The following lemma is about an arbitrary finite simple graph. In the application its vertices are geometric edges and its adjacency relation is proper crossing. Thus its conclusion includes interactions between the probes themselves.

Proposition 2.5 (Probe freezing)**.** *Let a finite simple graph have vertex partition X, T, Z, with induced subgraph on X equal to $J_{m,n}$. The objective is the number of monochromatic graph edges, and all vertices may be coloured freely. Use $V = mn$ and $\rho = 1/m + 1/n$. Suppose every vertex of X has at most αV neighbours outside X, every vertex of T has at most βV neighbours outside X, and*

$$\alpha, \beta \ge 0, \qquad \alpha + \rho \le a \le 2^{-4096}.$$

For $p \in T$ and $\sigma \in \{+, -\}$ let

$$h_p^\sigma = \sum_{x \in N(p) \cap X} s_x^\sigma.$$

This is its colour-1 cost minus its colour-0 cost when the grid is canonical. Choose $y_p^+ = 1$ when $h_p^+ < 0$ and $y_p^+ = 0$ otherwise, and put $y_p^- = 1 - y_p^+$. These are best responses in both phases, including their complemented zero-field tie convention.

Put $b = 104a + \beta$, and suppose that $Q \ge 0$ satisfies

$$|\{p \in T : |h_p^\sigma| \le bV\}| \le Q(b + \rho)V \quad \text{for both phases.} \tag{2.21}$$

Then every optimal colouring χ has a phase σ and a colouring ψ such that only vertices of T are changed, every probe is coloured y_p^σ, and

$$0 \le \Phi(\psi) - \Phi(\chi) \le E_{\mathrm{fr}} := 2Q(b + \rho)bV^2. \tag{2.22}$$

The unchanged grid differs from c^σ on at most $52aV$ vertices, all in $D_{220a} \cup C_{220a}$. If $\mathrm{OPT}_{\mathrm{fr}}$ is the minimum over colourings with the probes in either of the two frozen phases, then

$$0 \le \mathrm{OPT}_{\mathrm{fr}} - \mathrm{OPT} \le E_{\mathrm{fr}}.$$

It suffices for (2.21) to have the stronger estimate $|\{p : |h_p^\sigma| \le tV\}| \le Q(t + \rho)V$ for $0 \le t \le t_0$, together with $b \le t_0$.

Proof. Fix the colours outside X in a genuine global optimum. Its restriction to X minimizes the resulting grid objective. Each conditional unary oscillation is at most the number of outside neighbours, hence at most αV. Theorem 2.4 gives a phase σ and a disagreement set F with $|F| \le 52aV$ and the stated support. Keep this grid colouring fixed for the rest of the proof.

For every probe, its field from X differs from h_p^σ by at most $2|F| \le 104aV$. Under any colouring of all other vertices, its field from neighbours outside X has absolute value at most βV. Thus the full field differs from h_p^σ by at most bV. If $|h_p^\sigma| > bV$, the sign of the full field is the same as the canonical sign, whatever the outside colours are. Single-vertex optimality of χ therefore forces this hard probe already to have colour y_p^σ.

Recolour each remaining, soft probe to y_p^σ, in any order. At every step the grid remains fixed. The outside field still has absolute value at most βV even after earlier probe changes, since all probe–probe neighbours were included in that degree bound. The cost increase for the current recolouring is at most the magnitude of the current full field, which is bounded by

$$|h_p^\sigma| + 104aV + \beta V \le 2bV.$$

There are at most $Q(b+\rho)V$ soft probes by (2.21). Summing the sequential cost changes gives the upper bound in (2.22). The lower bound follows because χ is a global optimum and ψ is a feasible colouring. Only probes were changed, so the same grid and its localization remain.

The frozen feasible set is a subset of the original feasible set and contains the constructed ψ. Its optimum is therefore between $\Phi(\chi)$ and $\Phi(\psi)$, proving the final optimum bound. □

The last clause of Theorem 2.4 can now be applied to this unchanged grid under the newly frozen unary costs. It applies because the support is localized, whether or not this grid minimizes the new conditional objective. This is the reason for retaining the arbitrary localized-colouring estimate in addition to the optimum bracket.

3 Geometric interfaces and effective coordinates

We construct three sets of points: originals, frames, and programs. Original–frame edges realize the comparison grid; program–frame edges are the probes to which Proposition 2.5 will apply. All other edges of the complete graph remain present.

3.1 Two convex chains

For rational $D \ge 10$, $F \ge 1$, and $0 < \kappa, \kappa_f \le 1/100$, set

$$P(y) = (-D - \kappa y^2, y), \qquad A(t) = (F + \kappa_f t^2, t), \qquad -1 < y, t < 1. \tag{3.1}$$

Choose originals $P_j = P(y_j)$ in increasing parameter order, and frames $A_i = A(t_i)$ in decreasing parameter order, with $0 \le j < o$ and $0 \le i < B$.

Lemma 3.1 (The base drawing)**.** *The base points are in general position. The edges A_iP_j induce the strict comparison grid:*

$$A_iP_j \sim A_kP_l \quad \Longleftrightarrow \quad (i < k \text{ and } j < l) \text{ or } (k < i \text{ and } l < j).$$

An edge within either chain does not cross an edge joining the chains.

Proof. For $a < b < c$, direct expansion gives

$$\begin{aligned}
\operatorname{or}(P(a), P(c), A(t)) &= -(c-a)[D + F + \kappa_f t^2 + \kappa((a+c)t - ac)] < 0,\\
\operatorname{or}(P(a), P(c), P(b)) &= -\kappa(c-a)(b-a)(c-b) < 0,\\
\operatorname{or}(A(a), A(c), P(y)) &= (c-a)[D + F + \kappa y^2 + \kappa_f((a+c)y - ac)] > 0,\\
\operatorname{or}(A(a), A(c), A(b)) &= \kappa_f(c-a)(b-a)(c-b) > 0.
\end{aligned}$$

The brackets are at least $D + F - 3\kappa$ and $D + F - 3\kappa_f$, respectively. These formulas cover every single-chain and mixed triple. With three points on one chain, the middle point lies strictly inside the triangle formed by the other two and any point on the other chain: the displayed chord test and the mixed tests for the two subpairs give its three side tests. This excludes an internal-chain edge crossing a joining edge. Two points on each chain form a convex quadrilateral, since the opposite chain lies strictly on one side of either same-chain chord. Its diagonals give the stated comparison relation, with the two parameter orders reversed. □

There are S early source endpoint clusters and S late clusters, each containing W consecutive originals. Between them lie $D_m = 512SW$ dummy originals, so $o = 514SW$. Each cluster is an equally spaced interval of parameter diameter τ around a macro parameter. Early, dummy, and late parameters lie, respectively, in $(-3/4, -5/8)$, $(-1/2, 1/2)$, and $(5/8, 3/4)$. A matching pairs the early and late clusters; Section 4 specifies it from the source permutation.

Every original chord has direction $(b-a)(-\kappa(a+b), 1)$, while an original–frame vector has horizontal component at least $D + F$ and vertical component of absolute value at most two. These direction cones are separated. Programs will lie in $x < -D/2$, $|y| < 2$. The preliminary interfaces below use small displacements and clone diameter τ; the effective construction then chooses both by explicit simultaneous bounds.

3.2 Ordinary programs

Take $8 \mid B$ and $t_i = 1 - 2(i+1)/(B+1)$. The artificial point $A_* = A(0)$ lies in the strict angular gap between the middle frame rows, as viewed from every original. At a macro point $p = P(a)$, an ordinary program has prototype

$$r = p - \mu_o(A_* - p), \qquad \mu_o > 0. \tag{3.2}$$

Lemma 3.2 (Ordinary interface)**.** *For sufficiently small μ_o, followed by sufficiently small τ, every clone p_s of the parent satisfies*

$$rA_k \sim p_s A_i \quad \Longleftrightarrow \quad \begin{cases} i < k, & k < B/2, \\ i > k, & k \geq B/2. \end{cases} \tag{3.3}$$

Outside the parent block, the program fan has the neighborhood of a virtual column inserted at that block. A core edge from a parent clone to a different macro or a dummy crosses exactly the first half of the program fans when its other endpoint is later, and exactly the last half when it is earlier. These tables hold on an open neighborhood of the prototype.

Proof. Put $\epsilon_i = \operatorname{sgn} \operatorname{or}(r, p_s, A_i)$. The strict central gap gives $\epsilon_i = 1$ on the first half and -1 on the last half, uniformly for all clones after choosing τ. Both endpoint tests for proper crossing are needed. When $i < k$, they require $\epsilon_i = \epsilon_k = 1$; when $i > k$, they require $\epsilon_i = \epsilon_k = -1$. For $i = k$

there is a common endpoint. Here r and p_s lie on the same strict side of each frame chord: on $|y| \le 2$ that chord has supporting coordinate

$$x = F + \kappa_f((t_i + t_k)y - t_i t_k) \ge F - 5\kappa_f > 0,$$

whereas both points lie in the left strip. This proves (3.3). In every nonparent column the nonzero macro determinants persist under displacement and cloning, giving the virtual-column prefix or suffix.

For the core table, let $w = P(b)$ with $b \ne a$. Then $\operatorname{or}(p, w, r) = -\mu_o \operatorname{or}(p, w, A_*)$, so r and every frame point lie on opposite sides of pw. The other test compares the signs of $\operatorname{or}(r, A_k, p)$ and $\operatorname{or}(r, A_k, w)$. The former is negative on the first half and positive on the last; at zero displacement the latter has the sign of $b - a$, and this strict sign persists. This gives the claimed halves for the actual clones. All signs have positive margins, hence hold on an open neighborhood. No half-frame assertion is made for a short edge inside the parent cluster; its exact unary cost is retained later. □

3.3 Rank programs

For a selected macro chord $P(a)P(b)$, $a < b$, write $d_0 = b - a$ and let a_s, b_t, $1 \le s, t \le W$, be its clone parameters. Choose

$$0 < \mu_r < \frac{1}{100W}, \qquad 0 < \tau \le \frac{\mu_r d_0}{100}, \qquad y_L = a + \mu_r d_0, \quad y_R = b - \mu_r d_0.$$

The supporting horizontal coordinate of an actual selected chord at height y is

$$T_{s,t}(y) = -D - \kappa((a_s + b_t)y - a_s b_t). \tag{3.4}$$

Lemma 3.3 (Rank interface)**.** *At each internal gap $1 \le \ell < W$ there are open regions for left and right programs. Every left program fan crosses precisely the selected edges with $s > \ell$; every right fan crosses precisely those with $t \le \ell$. The test is independent of the opposite clone rank.*

The grid neighborhood is the exact virtual column after the entire early parent block, or before the entire late parent block, independently of ℓ. On an incident nonselected core edge to $P(w)$, every fan crosses if $a < w < b$, and none crosses otherwise.

Proof. At the unsteered center $r_0 = (1 - \theta)P(a) + \theta P(b)$, where $\theta = \mu_r$ or $1 - \mu_r$,

$$\operatorname{or}(r_0, A, P(a)) = \theta \operatorname{or}(P(a), P(b), A) < 0, \quad \operatorname{or}(r_0, A, P(b)) = (1 - \theta) \operatorname{or}(P(a), A, P(b)) > 0.$$

Cloning and sufficiently small horizontal steering preserve these opposite endpoint signs for every frame point. The remaining crossing test is exactly

$$rA \sim P(a_s)P(b_t) \quad \Longleftrightarrow \quad x_r < T_{s,t}(y_r). \tag{3.5}$$

Thus this is an internal segment crossing, not merely a support-line intersection.

At y_L, the support increases in s and decreases in t. The full ranges over the opposite rank satisfy

$$\begin{aligned} G_{L,\ell} &= T_{\ell+1,W}(y_L) - T_{\ell,1}(y_L) \\ &= \kappa[(a_{\ell+1} - a_\ell)(b_W - y_L) - (y_L - a_\ell)(b_W - b_1)] \ge \frac{\kappa\tau d_0}{2W} > 0. \end{aligned} \tag{3.6}$$

Indeed $a_{\ell+1} - a_\ell \geq \tau/W$, $b_W - y_L \geq 0.98 d_0$, and $y_L - a_\ell \leq 1.005\mu_r d_0 < 0.011 d_0/W$. Thus choose

$$x_{L,\ell} = \tfrac{1}{2}(T_{\ell,1}(y_L) + T_{\ell+1,W}(y_L)).$$

At y_R, reflection and rank reversal give $T_{1,\ell}(y_R) - T_{W,\ell+1}(y_R) \geq \kappa\tau d_0/(2W)$, so

$$x_{R,\ell} = \tfrac{1}{2}(T_{W,\ell+1}(y_R) + T_{1,\ell}(y_R))$$

gives exactly $t \leq \ell$. These prototypes tend to the unsteered centers as $\tau \to 0$, retaining the endpoint tests above.

For grid neighbors, marker-to-parent directions remain in the original-chord cone, outside the entire frame cone. The sign ϵ_i in the preceding proof is now constant over frame rows, positive for an original before the marker and negative for one after it. Consequently

$$rA_k \sim P_j A_i \quad \Longleftrightarrow \quad \begin{cases} i < k, & P_j \text{ precedes the marker,} \\ i > k, & P_j \text{ follows it.} \end{cases}$$

The left marker is above its whole parent block and the right marker below its whole parent block. Their displacement remains below the next macro gap, and steering does not change these signs.

For a nonselected incident chord, put $L_{a,w}(y) = -D - \kappa((a+w)y - aw)$. At the left unsteered center,

$$(r_0)_x - L_{a,w}((r_0)_y) = -\kappa\mu_r(b-a)(b-w).$$

The support-side test excludes $w > b$. For $w < b$, the other endpoint test excludes $w < a$, since its sign at $P(w)$ tends to the sign of $w - a$, while the near-endpoint sign is negative. Thus precisely $a < w < b$ remains. The right support difference is $\kappa\mu_r(b-a)(a-w)$; its near sign is positive and the other endpoint sign is negative precisely for $w < b$, giving the same interval. Strict macro margins preserve these statements. Parent-cluster short edges and the selected partner are excluded from this last table. □

Lemma 3.4 (Nonincident exclusion)**.** *A program fan does not cross a core edge whose endpoints both avoid its parent block.*

Proof. For endpoints in different nonparent blocks, the fan at the parent macro is noncrossing by Lemma 3.1, with nonzero macro determinants. Small displacement and then cloning preserve the exclusion. For a short edge in one nonparent block, the macro fan has positive distance from that block's center. Choose the displacement and clone neighborhood below this distance. The whole short edge remains disjoint from the fan. This argument avoids applying a strict determinant test to a collapsed zero-length edge. □

3.4 Canonical response fields

Let $N_g = Bo$, $\rho = 1/B + 1/o$, $u_k = (k + 1/2)/B$, and $v_j = (j + 1/2)/o$. The canonical grid spins are $\sigma_{kj} = 2\mathbf{1}_{u_k + v_j < 1} - 1$. The field of a fan is its grid cost in colour one minus its grid cost in colour zero. Its canonical response minimizes this cost, with colour zero at a tie. For a virtual column after r_0 originals, put $v = r_0/o$; its exact field is

$$\begin{aligned} H_k &= \sum_{j<r_0} \sum_{i<k} \sigma_{ij} + \sum_{j \geq r_0} \sum_{i>k} \sigma_{ij}, \\ H_{k+1} - H_k &= \sum_{j<r_0} \sigma_{kj} - \sum_{j \geq r_0} \sigma_{k+1,j}. \end{aligned} \tag{3.7}$$

There is no contribution from row k, whose fan shares a frame endpoint.

Lemma 3.5 (Field and soft-row bounds)**.** *Write $s_k = 1 - u_k - v$. Then*

$$\begin{aligned} |H_{k+1} - H_k - o(1 - 2|s_k|)| &\le 2 + 2o/B, \\ |H_k/N_g + s_k(1 - |s_k|)| &\le 6\rho. \end{aligned} \tag{3.8}$$

A rank program has field $g_k = H_k$. For an ordinary program, choose a virtual position at a parent-block boundary. Then $|g_k - H_k| \le 2BW$, and away from the central ordinary switch the increment error is at most $2W$.

Suppose $B \ge o \ge 1000$, $8 \mid B$, and $W/o \le 1/514$. For either program type and $0 \le z \le 1/100$,

$$|\{k : |g_k| \le zBo\}| \le 12Bz + 3. \tag{3.9}$$

An early program has response spin $+1$ on the middle $3B/4$ rows; a late program has spin -1 there. Each early half-frame sum is at least $B/4$ and its full sum at least $B/2$; the late sums have the opposite bounds. For s programs the total soft count is at most $Q(z + \rho)N_g$, where $Q = 12s/o$.

Proof. For $f_u(y) = 2\mathbf{1}_{y<1-u} - 1$, any midpoint sum over a prefix or suffix of column cells differs from o times the integral by at most one: only the threshold cell matters, including a midpoint tie. Also changing u by $1/B$ changes a suffix integral by at most $2/B$. Since

$$\int_0^v f_u(y)\,dy - \int_v^1 f_u(y)\,dy = 1 - 2|1 - u - v|,$$

the first estimate follows from (3.7). The two rectangles $[0, u_k] \times [0, v]$ and $[u_k, 1] \times [v, 1]$ differ from the actual cell union only in row k, of area $1/B$. The cellwise canonical function differs from $2\mathbf{1}_{x+y<1} - 1$ in L^1 by at most 2ρ. Integrating the latter function over the rectangles gives $-s_k(1 - |s_k|)$, proving even the bound 3ρ. Only the W parent columns differ for an ordinary program. Each prefix or suffix has absolute sum at most B and changes by at most one away from its switch. This proves its two error bounds.

For early programs $v \le 1/514$, and for late programs $v \ge 1 - 1/514$. A soft row satisfies

$$|s_k|(1 - |s_k|) \le z + 2W/o + 6\rho < 0.042 < 7/64.$$

It therefore belongs to one of $|s_k| \le 1/8$, $s_k \ge 7/8$, or $s_k \le -7/8$. For early ranks these lie in $u \le 1/8$ or $u \ge 3/4$; for late ranks in $u \le 1/4$ or $u \ge 7/8$. Each lies away from the ordinary switch. On each interval the main increment has constant sign and magnitude at least $3o/4$. Even the weaker error $6 + 2W$ leaves magnitude at least $o/2$. A field interval of length $2zBo$ contains at most $4Bz + 1$ such rows, proving (3.9).

On $u_k \in [1/8, 7/8]$, an early rank has $1/9 < s_k < 8/9$, so $s_k(1 - s_k) \ge 8/81$. The error $2/514 + 12/o < 0.016$ cannot change the negative field sign, hence the response colour is one. The late case reverses the sign. Each half-frame has $3B/8$ controlled and at most $B/8$ other rows, giving the claimed sums. Finally $12sBz + 3s \le (12s/o)(z + \rho)N_g$. □

A frozen fan of colour c contributes $2c - 1$ to the core unary coefficient. Early ordinary halves thus contribute positively, and late halves negatively. Early rank programs add positive suffix steps; late ones add negative prefixes whose disappearance is a positive step. Their full weights lie in $[B/2, B] \subseteq [B/4, B]$. All internal gaps at an endpoint have the same response sequence; copies in one open prototype region retain it, independently of other programs.

3.5 All remaining edge classes

Partition the complete graph into X (original–frame), T_f (program–frame), C (original–original), Y (program–original or program–program), and I (frame–frame).

Lemma 3.6 (Convex separation)**.** *The programs can all be placed strictly inside $\mathcal{H} = \{(x, y) : \phi(x, y) \le 0\}$, where $\phi(x, y) = x + D + \kappa y^2$. Every open X segment lies outside $\mathcal{H}$. Consequently X has no crossing with C or Y. The frame class I has no crossing-graph neighbor outside itself.*

Proof. The set $\mathcal{H}$ is convex. For $z_\lambda = (1 - \lambda)P(a) + \lambda A(t)$,

$$\phi(z_\lambda) = \lambda[D + F + \kappa_f t^2 + \kappa(2at - a^2)] + \kappa\lambda^2(t - a)^2. \tag{3.10}$$

Its bracket is positive, so every open grid segment is outside. Substituting $\lambda = -\mu_o$ gives, for $0 < \mu_o \le 1$, $\phi(r) \le -\mu_o(D + F - 7\kappa) < 0$ at an ordinary prototype. At a rank height, $a_s < y_r < b_t$ and

$$\phi(T_{s,t}(y_r), y_r) = -\kappa(y_r - a_s)(b_t - y_r) < 0.$$

Its steered horizontal coordinate is a midpoint of two such supports; ϕ is affine in x, so the actual prototype is inside. If $\eta_{\text{cap}} = \min_r(-\phi(r)) > 0$, perturbation by at most $\delta \le \min(1/4, \eta_{\text{cap}}/8)$ in both coordinates changes ϕ by less than 2δ on $|y| \le 2$. Hence every final program remains inside. All C, Y endpoints lie in $\mathcal{H}$, so convexity excludes their crossing with X.

For the frame component put $\psi(x, y) = F + \kappa_f y^2 - x$ and use $\mathcal{H}_f = \{(x, y) : \psi(x, y) \le 0\}$. It contains all I segments. Every nonframe point r has $x_r \le -D$, $|y_r| \le 2$, and along a frame fan

$$\psi((1 - \lambda)A(t) + \lambda r) = \lambda[F - x_r + \kappa_f(2ty_r - t^2)] + \kappa_f\lambda^2(y_r - t)^2 > 0$$

for $0 < \lambda \le 1$, since the bracket is at least $D + F - 5\kappa_f$. Thus no fan crosses I. The remaining C, Y segments lie in $x \le -D$, disjoint from $\mathcal{H}_f \subseteq \{x \ge F\}$. □

The isolated component I has optimum $Z(B)$, not zero. Besides I–I, the only possible crossing types are

$$X{:}X, \quad X{:}T_f, \quad T_f{:}T_f, \quad C{:}C, \quad C{:}T_f, \quad Y{:}T_f, \quad Y{:}C, \quad Y{:}Y.$$

All interactions involving multiple programs are retained. Let G_0 be the crossing graph after deleting Y, and set

$$p = Bs, \qquad c = \binom{o}{2}, \qquad h = os + \binom{s}{2}, \qquad R_Y = h(p + c) + \binom{h}{2}. \tag{3.11}$$

Deletion removes only the last three crossing types, so

$$\mathrm{OPT}(G_0) \le \mathrm{OPT}(P) \le \mathrm{OPT}(G_0) + R_Y. \tag{3.12}$$

The upper bound extends any colouring of G_0 by arbitrary colours on Y, charging at most one per deleted constraint.

3.6 A deterministic rational realization

We now give explicit rational scales that instantiate all preceding open-region statements. These are the scales used in the formal development.

Fix $D = 10$, $F = 1$, and $\kappa = \kappa_f = 1/100$. Having fixed the integer counts and source pairing, choose the early macros, physically ordered late macros, dummies, and frames by

$$a_j = -\tfrac{3}{4} + \frac{j}{8(S+1)}, \quad b_l = \tfrac{5}{8} + \frac{l}{8(S+1)}, \quad z_v = -\tfrac{1}{4} + \frac{v}{2(D_m+1)}, \quad t_i = 1 - \frac{2(i+1)}{B+1}. \tag{3.13}$$

Here $1 \le j, l \le S$, $1 \le v \le D_m$, and $0 \le i < B$. Let $\mathcal{A}$ be the set of early and late macro parameters and all dummy parameters; frame parameters are not included. Define

$$\eta = \min(\{1\} \cup \{|a-c| : a, c \in \mathcal{A},\ a \ne c\}), \quad \mu = \frac{\eta^2}{10000(W+1)}, \quad \tau = \frac{\mu\eta^2}{1000(B+1)}. \tag{3.14}$$

Use $\mu_o = \mu_r = \mu$ and clone parameters $a_r = a + \tau\vartheta_r$, where $\vartheta_r = (2r - W - 1)/(2(W-1))$. All internal rank prototypes use the midpoint formulas of Lemma 3.3; there are $2S(W-1)$ candidates, including those assigned zero copies later.

Lemma 3.7 (Simultaneous quantitative margins)**.** *The choices* (3.14) *satisfy all geometric interfaces above for the actual clones and every candidate program. In particular no choice depends on later program multiplicities.*

Proof. The set $\mathcal{A}$ is finite with distinct parameters, so $0 < \eta \le 1$. Direct substitution gives

$$\begin{aligned} &0 < \mu \le 1/10000, \quad 100W\mu < 1, \quad 10000\mu \le \eta^2, \quad 4800\mu < \eta^2, \\ &0 < \tau \le \eta/1000, \quad 1000\tau \le \mu\eta^2, \quad 3(B+1)\tau/2 \le \mu, \quad 100\tau \le \mu(b-a) \end{aligned} \tag{3.15}$$

for any selected $a < b$, since $b - a \ge \eta$. Every clone remains in $(-1, 1)$ and within $\tau/2$ of its macro. A nonparent original parameter z remains on the same side of the parent macro a as its own macro and obeys $|z - a| \ge \eta/2$. Consequently

$$5\mu \le |z-a|, \qquad 2\mu(b-a) \le |z-a|. \tag{3.16}$$

If z also avoids the selected partner macro b, then $|b - z| \ge \eta/2$ and

$$20\tau \le \mu(b-a)|b-z|. \tag{3.17}$$

For two nonparent actual parameters z, v, even when they are clones of the same macro, their distances to a are each at least $\eta/2$. Thus

$$1200\mu < |z-a||v-a|, \qquad 200\mu(b-a) < |z-a||v-a|. \tag{3.18}$$

The second inequality uses $b - a \le 2$.

We check that these numerical inequalities pay for both segment-side tests, rather than only support-line intersections. In the ordinary own-column test, the unperturbed parent determinant has magnitude at least $10\mu|t_i| \ge 10\mu/(B+1)$. Moving its original parameter from a to a clone c changes it by at most $25|c-a|$. The bound $3(B+1)|c-a| \le \mu$ therefore preserves its sign. The constant 25 follows by expanding the determinant difference:

$$\mathrm{or}(r, P(c), A(t)) - \mathrm{or}(r, P(a), A(t)) = (c-a)(r_x - A(t)_x - (a+c)(t - r_y)/100);$$

here $|r_x| \le 22$, $|r_y| \le 2$. For a nonparent ray, ordinary displacement changes the determinant by at most 36μ, while its macro magnitude is at least $10|z-a|$. Inequality (3.16) preserves that sign. These bounds establish the full ordinary grid table, including its central switch.

For a core support $L_{c,z}(y) = -10 - ((c+z)y - cz)/100$, the ordinary offset is exactly

$$r_x - L_{c,z}(r_y) = -\frac{(a-c)(a-z)}{100} - \mu C(a,c,z), \qquad C(a,c,z) = 11 + \frac{a^2 - ac - az}{100},$$

with $10 \le C(a,c,z) \le 12$ for parameters in $[-1,1]$. When c is a parent clone, the first term has absolute value at most $\tau/100$; the second is strictly negative and dominates it. Combining with the parent and nonparent ray signs gives the ordinary core half-frame table. For two nonparent endpoints, the product margin (3.18) preserves the macro support sign when the parent lies between them; if they are on one side, both ray signs agree. In either case there is no crossing. This also proves nonincidence for a short edge in a different cluster without taking a zero-length limit.

For rank programs, the last inequality of (3.15) and $100W\mu < 1$ are exactly the full-range gap hypotheses in (3.6). The midpoint has horizontal steering from its unsteered macro-chord support at most $\tau/50$, obtained by expanding (3.4) with endpoint deviations at most $\tau/2$. Its L^∞ displacement from its parent is at most $\mu(b-a)$. For either actual selected endpoint c, the height gap is at least τ and

$$|x_r - P(c)_x| \le |y_r - c|/10.$$

Indeed for a support through parameters u, v,

$$L_{u,v}(y) - P(c)_x = ((c-u)(c-v) - (u+v)(y-c))/100;$$

one endpoint differs from c by at most τ, $|c-u|, |c-v|, |u+v| \le 2$, and averaging preserves the bound. The horizontal distance to every frame point is at least 11, while its vertical distance is at most two. The endpoint determinant thus has the strict sign prescribed by whether $c < y_r$ or $c > y_r$. This proves the endpoint-side test in (3.5) uniformly over both ranks.

A rank displacement bounded by $\delta_r = \mu(b-a)$ changes a nonparent ray determinant by at most $14\delta_r$, compared with macro magnitude at least $10|z-a|$; use (3.16). For a nonselected incident edge from parent clone c to actual z, the left support offset has the explicit error bound

$$\left| r_x - L_{c,z}(r_y) + \frac{\mu(b-a)(b-z)}{100} \right| \le \frac{3\tau}{100}.$$

It is the steering error $\tau/50$ plus clone error at most $\tau/100$. Inequality (3.17) preserves its sign, so the ray and support tests together give exactly $a < z < b$. The right statement follows by reflection. Finally any rank displacement of size δ_r changes a nonincident chord offset by at most $2\delta_r$, whereas its initial curvature offset is $|(a-z)(a-v)|/100$. Inequality (3.18) again preserves the sign. Endpoints on the same side give equal ray signs, and endpoints straddling the parent put program and frame on the same side of their chord. This proves rank nonincidence in all cases.

The strict inwardness formulas in Lemma 3.6 apply directly to these ordinary and steered rank prototypes. Their heights have absolute value below two. Thus all prototypes are valid simultaneously. The argument uses only fixed base data and the candidate gap index. □

Compute each candidate's exact grid neighbors and response sequence. These determine the ordinary and rank weights. The greedy and ceiling rules of Section 4 now determine the actual multiplicities and hence s. None of the earlier choices is repeated.

Let $\mathcal{B}$ be the set of all fixed original and frame points, and $\mathcal{R}$ the finite set of used prototypes.

Define positive rational margins

$$\begin{aligned}
\eta_{\mathrm{or}} &= \min\bigl(\{1\} \cup \{|\operatorname{or}(u,v,r)| : u,v \in \mathcal{B},\ r \in \mathcal{R},\ \operatorname{or}(u,v,r) \neq 0\}\bigr),\\
\eta_{\mathrm{cap}} &= \min\bigl(\{1\} \cup \{-\phi(r) : r \in \mathcal{R}\}\bigr),\\
\eta_{\mathrm{ht}} &= \min\bigl(\{1\} \cup \{2 - |r_y| : r \in \mathcal{R}\}\bigr),\\
L_{\mathcal{B}} &= 1 + 2\sum_{u \in \mathcal{B}} (|u_x| + |u_y|),\\
\delta &= \min\{1/4, \eta_{\mathrm{cap}}/8, \eta_{\mathrm{ht}}, \eta_{\mathrm{or}}/(8L_{\mathcal{B}})\}.
\end{aligned}$$

These are the finite-minimum radius formulas used in the formalization. The value one handles empty sets. Every nonzero one-program determinant retains its sign under a coordinate change at most δ: its change is bounded by $\delta(|u_x - v_x| + |u_y - v_y|) < L_{\mathcal{B}}\delta \leq \eta_{\mathrm{or}}/8$. At the insertion points below the displacement is less than $\delta/2$ per coordinate. Hence the cap and height bounds are preserved as well.

An unused zero determinant need not retain its value. Every prescribed crossing has four strict determinant tests. A pair sharing a base endpoint remains noncrossing by definition; each other prescribed exclusion has either a strict same-side support pair or strict same-side fan rays. The quantitative proof above supplies these certificates, including parent-cluster short edges, whose two fan rays have the same nonzero sign. Thus preserving the nonzero determinants suffices. Let $N = B + o + s$ be the final total point count. Set $m = N^2 + N + 2$. For each required copy of prototype r, test in lexicographic order the m^2 rational candidates

$$r + \delta\left(\frac{i}{m+1} - \frac{1}{2}, \frac{j}{m+1} - \frac{1}{2}\right), \qquad 1 \leq i, j \leq m. \tag{3.19}$$

Reject coincidences with existing points and collinearity with any existing pair. At most $\binom{N}{2}$ forbidden lines each meet this grid in at most m points, and at most N further points are forbidden. Since $\binom{N}{2}m + N < m^2$, some candidate survives. Induction gives distinct points in general position. The same radius and denominator are used throughout. All required one-program tables and response colours persist; actual fan–fan crossings and short-edge unary costs are evaluated after insertion, while program nonfans are covered by (3.11).

Proposition 3.8 (Effective realization)**.** *The construction is a deterministic polynomial-time algorithm with rational coordinates of polynomial binary length. It preserves all stated interfaces simultaneously and yields general position. A common positive scaling yields integer coordinates of polynomial binary length with the same crossing relation.*

Proof. Proposition 4.8 bounds the actual program count, and $2S(W-1)$ bounds the candidate prototypes. Both, together with B, o, are fixed polynomials in the source size. The base coordinates have $O(\log(B+1)+\log(W+1)+\log(S+1)+\log(D_m+1)+1)$ bits. The fixed formulas for η, μ, τ use a bounded number of rational operations on base parameters and counts. Prototype coordinates are bounded-degree expressions in these values. Each determinant margin therefore has polynomial bit length; finite minima select existing values. The sum defining $L_{\mathcal{B}}$ has polynomially many terms, so its denominator bit length is bounded by the sum of their polynomial bit lengths. The same holds for δ. Internal rank gaps use direct formulas, not a chain of nested scales.

The integer greedy recurrence is evaluated with cached cumulative weights. Inserting at most N programs tests $O(N^4)$ candidates per insertion, each against $O(N^2)$ exact equations and determinants, for $O(N^7)$ rational operations. All intermediate bit lengths are polynomial; standard integer arithmetic therefore implements the whole algorithm in polynomial bit time. Earlier inserted points

only reject candidates and never redefine later scales. Finally the product of the positive coordinate denominators has bit length at most their sum. Scaling by this product preserves strict determinant signs and gives integer coordinates with polynomial bit length. □

4 Encoding the source graph in the core

Throughout this section the program fans T_f are fixed to their canonical best responses t^+ to the grid. We prove a statement about the entire original–original subsystem with these frozen neighbors. Colours forced here are consequences of minimization in this subsystem; they are never imposed in the target instance.

4.1 Source matching and amplification

Let G be a permutation graph on n vertices, with a supplied permutation representation, and write

$$g(G) = \min_{z:V(G)\to\{0,1\}} \sum_{uv\in E(G)} \mathbf{1}_{z_u=z_v} = |E(G)| - \mathrm{MaxCut}(G).$$

This represented source is NP-hard by the construction in [2]. Place its two endpoint orders in consecutive blocks of the original cap, reversing one order. The resulting chords alternate exactly for adjacent source vertices. Replace each source chord by q macro copies, ordered forward at the early endpoint and backward at the late endpoint. Copies of a vertex are nested; two distinct source classes have either all q^2 crossings or none. Write $S = nq$ and label the selected macro chords p_jq_j, $1 \le j \le S$, by early endpoint order.

Each macro endpoint is replaced by W consecutive originals, denoted $p_{j,r}, q_{j,t}$, $1 \le r, t \le W$. There are $D_m = 512SW$ dummy points between the endpoint blocks. All original points are in convex position on the same cap, so their chords cross precisely when their endpoints alternate.

Partition the core C into selected edges

$$\mathcal{S} = \bigcup_{j=1}^{S} \mathcal{S}_j, \qquad \mathcal{S}_j = \{p_{j,r}q_{j,t} : 1 \le r, t \le W\},$$

short edges $\mathcal{U}$ within one endpoint cluster, dummy–dummy edges $\mathcal{D}$, and the remaining palette edges $\mathcal{F}$. Define the proposed palette by

$$f^*(e) = \begin{cases} 0, & e \text{ joins two early clusters or an early cluster to a dummy,} \\ 1, & e \text{ joins two late clusters or a dummy to a late cluster,} \\ \mathbf{1}_{j<k}, & e = p_{j,r}q_{k,t},\ j \ne k. \end{cases} \tag{4.1}$$

Every palette edge falls in exactly one case.

Lemma 4.1 (Independent source amplification)**.** *After deleting the crossing constraints internal to each $\mathcal{S}_j$, the graph induced on $\mathcal{S}$ is the independent qW^2-fold blowup of G. Its minimum monochromatic count is $\Delta g(G)$, where $\Delta = q^2W^4$.*

Proof. Consecutive endpoint clusters preserve all crossings between different macro chords. The q macro copies of one vertex remain pairwise noncrossing, and the deletion makes each individual bundle independent. Thus each source vertex becomes an independent class of qW^2 variables with identical neighbors; source adjacency gives a complete join.

With colours outside one class fixed, its cost is affine in the number of colour-one variables in that class. One uniform colour is therefore no worse. Perform this operation successively on all classes. The final class-uniform colouring costs $(qW^2)^2$ times a colouring cost of G. Conversely every source colouring lifts to that cost. The deleted internal constraints remain in the actual drawing and are restored below. □

4.2 The frozen interface and affine fields

Every unary coefficient means cost in colour one minus cost in colour zero; a frozen neighbor of colour a contributes $2a-1$. With $e \sim f$ denoting proper crossing, define

$$E_C(x;t^+) = \sum_{\substack{\{e,f\}\subseteq C\\ e\sim f}} \mathbf{1}_{x_e=x_f} + \sum_{\substack{e\in C,\ f\in T_f\\ e\sim f}} \mathbf{1}_{x_e=t_f^+}, \qquad M_C = \min_{x:C\to\{0,1\}} E_C(x;t^+). \tag{4.2}$$

The first sum ranges over unordered pairs of distinct edges.

By Lemmas 3.2–3.5, an ordinary program at an early endpoint contributes a positive half-frame weight in $[B/4, B]$ on an incident nonshort core edge; at a late endpoint the weight is negative with magnitude in that interval. Which half is used depends only on endpoint order, not clone ranks. An early rank program at gap ℓ adds a positive weight $w_\ell \in [B/4, B]$ on the suffix $r > \ell$. A late program adds $-w_\ell$ on the prefix $t \le \ell$. No program fan meets a nonincident core edge. Its contribution on any incident edge is bounded in magnitude by B. The actual unary on a short edge is computed directly. Response weights depend only on one prototype and the base grid; their computation is independent of other multiplicities and core colours.

Lemma 4.2 (Affine palette field)**.** *The palette contribution to a selected edge is*

$$\sum_{\substack{f\in\mathcal{F}\\ f\sim p_{j,r}q_{j,t}}} (2f^*(f)-1) = \beta_j + \gamma_{j,L}(r-1) + \gamma_{j,R}(t-1), \tag{4.3}$$

where the integer coefficients are computable by crossing counts and obey $|\beta_j| \le o^2$ *and* $-o \le \gamma_{j,L}, \gamma_{j,R} < 0$.

Proof. A palette edge avoiding both parent clusters contributes independently of r, t. One with an endpoint in the early parent crosses according to whether that endpoint precedes or follows r. Its other endpoint avoids the late parent, and its colour is constant over the early cluster; its sum is affine in r alone. The late cluster similarly gives a function of t. An edge between the two parent clusters is selected, and one inside a cluster is short, so neither contributes a mixed term to the palette field.

The dummy neighbors give exactly

$$-D_m(r-1) + D_m(W-t).$$

The early–dummy edges have colour zero and cross when their parent rank is below r; the dummy–late edges have colour one and cross when their parent rank is above t. Changing either rank changes at most one contribution for each nondummy endpoint, of which there are at most $2SW$. Thus either slope lies in $[-D_m - 2SW, -D_m + 2SW] \subseteq [-o, 0)$. Finally the number of possible palette neighbors is less than o^2, proving the intercept bound. □

Lemma 4.3 (Greedy rank correction)**.** *Each slope in Lemma 4.2 can be corrected with at most* $4oW/B + 4$ *rank programs at that endpoint. After correction the field on bundle* j *is*

$$b_j + \varepsilon_{j,L}(r) + \varepsilon_{j,R}(t), \qquad 0 \le \varepsilon_{j,L}(r), \varepsilon_{j,R}(t) < B, \qquad |b_j| \le 4o^2. \tag{4.4}$$

Proof. For $\gamma < 0$, let $f(r) = -\gamma(r-1)$, so $f(1) = 0$ and $f(W) \le oW$. Starting with $R(1) = 0$, at gap $r-1$ add the least nonnegative number of its weight-w_{r-1} programs that makes the new cumulative weight $R(r)$ at least $f(r)$. Inductively

$$f(r) \le R(r) < f(r) + B. \tag{4.5}$$

If copies are added, the final overshoot is smaller than the last weight, at most B. If none is added, the preceding overshoot remains below B because the target increased. Each copy weighs at least $B/4$, while $R(W) < oW + B$, giving the stated count. Unoccupied gaps add no copies; there is no extra term proportional to W.

At the early endpoint the coefficient is $R(r)$, which cancels the slope with error in $[0, B)$. At the late endpoint the negative prefixes give $-R(W) + R(t)$. Their initial offset must be retained: $b_j = \beta_j - R_{j,R}(W)$. Since $W \le o$ and $B \le o^2/1024$, $|b_j| \le o^2 + oW + B \le 4o^2$. Only the $W-1$ internal gaps are used. □

4.3 Forcing the palette inside the core

Use

$$T = 2^{13}o^2, \qquad A_0 = 2^{13}(S+1)T, \qquad \theta_j = jT.$$

For early–late edges let $u_{L,j}$ be the ordinary positive half-frame weight at early endpoint j, and $u_{R,j}$ the magnitude of the negative weight at its late partner. Choose

$$\begin{aligned} C_{L,j} &= \left\lceil \frac{A_0 + \theta_j - b_j}{u_{L,j}} \right\rceil, & C_{L,j}u_{L,j} &= A_0 + \theta_j - b_j + e_{L,j}, \\ C_{R,j} &= \left\lceil \frac{A_0 + \theta_j}{u_{R,j}} \right\rceil, & C_{R,j}u_{R,j} &= A_0 + \theta_j + e_{R,j}, \end{aligned} \tag{4.6}$$

where $0 \le e_{L,j}, e_{R,j} < B$. Both targets are positive. These ceilings use only known integer weights and palette crossing counts.

Lemma 4.4 (Palette forcing and selected residuals)**.** *Every minimizer defining M_C agrees with f^* on $\mathcal{F}$. Once the palette and fans are fixed, each selected edge has unary coefficient r_e with $|r_e| < 4B$; in particular $|r_e| \le 10B$.*

Proof. For a nonselected early-j–late-k edge, $j \ne k$, the ordinary coefficient is

$$\theta_j - \theta_k - b_j + e_{L,j} - e_{R,k}.$$

The first term has absolute value at least T. All adverse terms, including arbitrary remaining core colours, have magnitude at most

$$\begin{aligned} |b_j| + 2B + 2(4oW/B + 4)B + \binom{o-2}{2} &\le \tfrac{9}{2}o^2 + 8oW + 10B \\ &< 5o^2 < T. \end{aligned}$$

The last strict bound uses $o \ge 514W$ and $B \le o^2/1024$. The rank term counts both incident endpoint families; nonincident programs contribute nothing. The binomial term bounds the full core crossing degree, without assuming any other palette colour. A wrong colour can therefore be improved by a single flip. The preferred colour is zero for $j > k$ and one for $j < k$.

For an edge on one side, the two ordinary fields have the same sign. Each ordinary multiplicity is at least $A_0/(2B)$ because its target is at least $A_0 - 4o^2$. Either half-frame then contributes magnitude at least $A_0/8$ from each parent. A side–dummy edge has one such contribution. All adverse rank and core terms have magnitude at most

$$8oW + 8B + \binom{o-2}{2} < o^2 < A_0/8.$$

Thus early-side edges prefer zero and late-side edges prefer one. This covers the palette simultaneously.

For the selected edge $p_{j,r}q_{j,t}$, the ordinary counts cancel b_j in (4.4), leaving exactly

$$r_e = \varepsilon_{j,L}(r) + \varepsilon_{j,R}(t) + e_{L,j} - e_{R,j}. \tag{4.7}$$

All four errors have magnitude below B, proving the residual bound. □

This lemma is restricted to (4.2). Before fan freezing, a global optimum need not have this palette. The full-drawing lower bound will use only that its post-freezing core energy is at least M_C.

4.4 Dummy edges and local constraints

Lemma 4.5 (Exact dummy elimination)**.** *With the palette fixed, the optimized dummy-edge contribution, including its palette interactions, is*

$$Z(D_m) + SW\binom{D_m}{3}, \tag{4.8}$$

independently of all selected and short-edge colours.

Proof. A dummy chord with ranks $a < b$ crosses no early–late selected chord and no endpoint-cluster short edge. Its palette neighbors are exactly the side–dummy edges whose dummy endpoint is strictly between a, b. It has $SW(b-a-1)$ such neighbors of each colour. Its palette cost is consequently the constant $SW(b-a-1)$ in either colour. There are no program-fan neighbors by Lemma 3.4. Summing over $a < b$ gives $SW\binom{D_m}{3}$. What remains is the full convex drawing on D_m points, with optimum $Z(D_m)$ by the two-page complete-graph theorem [1]. Its endpoint-alternation predicate agrees with that of convex straight-line chords. □

Let Q_{loc} be the crossing constraints internal to one selected bundle, between selected and short edges, and between two short edges. These are crossing-graph constraints, not removed geometric edges.

Lemma 4.6 (Conservative local budget)**.**

$$|Q_{\mathrm{loc}}| \le 2SW^4. \tag{4.9}$$

After deleting these constraints and fixing the palette, every short edge has only unary costs. The dummy subsystem remains independent, and the remaining selected-edge graph is that of Lemma 4.1.

Proof. In one selected bundle, an internal crossing is specified by two early and two late ranks, giving at most $W^4/4$ choices. For a source–short crossing, the short chord belongs to one of that source's two endpoint clusters and its endpoints surround the source endpoint. An upper bound, even without this surrounding condition, is $2\binom{W}{2}W^2 \le W^4$ per bundle. A short–short crossing must lie in one cluster; the two clusters contribute at most $2\binom{W}{2}^2 \le W^4/2$. The sum is at most $7W^4/4 \le 2W^4$.

A short edge cannot cross a selected bundle avoiding its cluster, since the cluster is consecutive in cap order; short edges in distinct clusters cannot alternate. Thus the listed types exhaust its unfixed neighbors, except for dummy edges, which are noncrossing with it by Lemma 4.5. The remaining fixed neighbors give unary costs. Removing internal bundle crossings leaves exactly the source graph already identified. □

4.5 A computable core baseline

Delete Q_{loc} from (4.2). Fix the palette, set all selected colours to zero, independently minimize each short unary, and insert the dummy optimum (4.8). Call this integer C_0, and define

$$K_C = C_0 - \Delta|E(G)| + \sum_{e\in\mathcal{S}} \min\{0, r_e\}. \tag{4.10}$$

All of these operations are explicit sums, crossing tests, and minima of two integers. In particular neither $g(G)$ nor M_C is used.

For a fixed selected colouring x, let $\Psi(x)$ be the minimum of the deleted energy over the short and dummy variables, with the palette fixed. The exact identity is

$$\begin{aligned}\Psi(x) &= C_0 - \Delta|E(G)| + \Phi_{\mathrm{main}}(x) + \sum_{e\in\mathcal{S}} r_e x_e \\ &= K_C + \Phi_{\mathrm{main}}(x) + \sum_{e\in\mathcal{S}} (r_e x_e - \min\{0, r_e\}).\end{aligned} \tag{4.11}$$

Indeed, the short minima and dummy value are independent of x after deletion. All remaining source-dependent fixed-neighbor costs have coefficient r_e. The all-zero selected colouring contributes exactly $\Delta|E(G)|$ to the main graph, explaining the single subtraction. No internal-bundle contribution is subtracted twice.

Proposition 4.7 (Core encoding)**.** *The baseline K_C is computable in polynomial time, and*

$$K_C + \Delta g(G) \le M_C \le K_C + \Delta g(G) + 10BSW^2 + 2SW^4. \tag{4.12}$$

Proof. Every core minimizer has the palette by Lemma 4.4. Deleting nonnegative local terms and applying (4.11) gives the lower bound, since $\Phi_{\mathrm{main}}(x) \ge \Delta g(G)$ and each residual excess is nonnegative.

For the upper bound take a class-uniform selected colouring attaining $\Delta g(G)$, independently minimize every short unary, and choose an optimal dummy colouring. These choices coexist in one complete core colouring. Each selected unary exceeds its separate minimum by at most $|r_e| \le 10B$, and $|\mathcal{S}| = SW^2$. Restoring all local constraints costs at most $2SW^4$ by Lemma 4.6. The formula (4.10) proves computability with polynomially many exact operations. The optimizers just used certify the inequalities and are not computed by the reduction. □

The $10BSW^2$ budget is deliberately generous. The exact residual (4.7) permits $4BSW^2$ in this unordered-edge presentation. The formal development's conservative cardinality bookkeeping yields $8SH^{13} \le 10SH^{13}$, which is the same stated budget because $B = H^5$ and $W = H^4$.

Proposition 4.8 (Program count)**.** *The number of required program points satisfies*

$$s \le \frac{16SA_0}{B} + \frac{8SoW}{B} + 10S \le 2^{80}(S+1)^4H^3. \tag{4.13}$$

In particular $s \le o$. All multiplicities are computable before final general-position insertion.

Proof. Each target in (4.6) is at most $A_0 + ST + 4o^2 < 2A_0$. A weight is at least $B/4$, so each endpoint uses at most $8A_0/B + 1$ ordinary programs. Lemma 4.3 uses at most $4oW/B + 4$ rank programs per endpoint. Multiplication by $2S$ gives the first bound.

For $d = S + 1$, $o = 514SH^4 < 2^{10}dH^4$ and $A_0 = 2^{26}do^2 < 2^{46}d^3H^8$. Thus the first two terms are below $2^{50}d^4H^3$ and $2^{13}d^2H^3$, respectively, and $10S$ is smaller still. Their sum is below $2^{80}d^4H^3$. Since $H = 2^{20000}d^{40}$, this is smaller than $514SH^4 = o$ for $S \ge 1$. Each step uses only the individually known response weights; placing the resulting number of copies in their open neighborhoods preserves those weights. □

5 The complete drawing and the decision reduction

All coordinates and crossing predicates in this section refer to the final general-position drawing from Proposition 3.8. Its total number of points is $N = B+o+s$; the comparison grid has $N_g = Bo$ vertices. The notation Φ_{AB} means the sum of monochromatic constraints with one endpoint in each of the distinct crossing-graph vertex classes A, B; Φ_A counts unordered pairs within A.

5.1 A common computable background

Let G_0 be the crossing graph with Y deleted. Its remaining constraints are precisely

$$X{:}X, \quad X{:}T_f, \quad T_f{:}T_f, \quad C{:}C, \quad C{:}T_f, \quad I{:}I.$$

Write c^+ for the canonical grid colouring and t^+ for the independently computed program-fan responses, with colour zero at a zero field. Define

$$\begin{aligned} E_X(x; t^+) &= \Phi_X(x) + \Phi_{XT_f}(x, t^+), \\ d_v &= E_X(c^+ \oplus \{v\}; t^+) - E_X(c^+; t^+), \\ L_+ &= E_X(c^+; t^+) + \sum_{v \in X} \min\{0, d_v\}, \\ K_0 &= Z(B) + \Phi_{T_f}(t^+). \end{aligned} \tag{5.1}$$

The fan–fan term is evaluated after insertion, including every crossing that changed during program placement. These constants require no optimal grid or fan colouring.

Proposition 5.1 (Decoupling with the grid fixed)**.** *Suppose the hypotheses of Theorem 2.4 and Proposition 2.5 hold with grid dimensions B, o, grid size $N_g = Bo$,*

$$\rho = 1/B + 1/o, \quad a = \alpha + \rho \le 2^{-4096}, \quad Q = 12s/o, \quad b = 104a + \beta \le 1/100.$$

Set $E_g = 2^{32}N_g^2a^3$ and $E_{\mathrm{fr}} = 2Q(b+\rho)bN_g^2$. Then

$$K_0 + L_+ + M_C - E_{\mathrm{fr}} - E_g \le \mathrm{OPT}(G_0) \le K_0 + L_+ + M_C + E_g, \tag{5.2}$$

$$K_0 + L_+ + M_C - E_{\mathrm{fr}} - E_g \le \mathrm{OPT}(P) \le K_0 + L_+ + M_C + E_g + R_Y. \tag{5.3}$$

Proof. Take an actual optimum of G_0. Conditional on every other colour, its grid minimizes a unary-perturbed comparison energy with unary differences bounded by αN_g. Theorem 2.4 supplies a canonical phase and a localized flip set of size at most $52aN_g$. If necessary complement every colour in the whole graph to make the phase positive. There are no prescribed colours, and complementation preserves every monochromatic count.

Keep this grid unchanged. By Lemma 3.5, the probe soft-count hypothesis holds. Proposition 2.5 changes only the program fans to t^+, with cost increase at most E_{fr}. The grid retains its localized flip set, so the localized lower estimate in Theorem 2.4, now for the frozen unary energy in (5.1), gives at least $L_+ - E_g$. The current core colouring costs at least M_C, irrespective of whether its palette was forced before the fans were frozen. The frame class costs at least $Z(B)$, and the fan–fan cost is exactly $\Phi_{T_f}(t^+)$. These terms exhaust G_0. Thus the altered colouring has cost at least $K_0 + L_+ + M_C - E_g$ but at most $\mathrm{OPT}(G_0) + E_{\mathrm{fr}}$, proving the lower bound.

For the upper bound, use the same t^+, the grid colouring $c^+ \oplus \{v : d_v < 0\}$, a core minimizer, and a convex frame optimum. The prescribed-phase upper witness in Theorem 2.4 has grid cost at most $L_+ + E_g$. All these choices coexist, since the edge classes are disjoint and no further cross-class constraints remain. This proves (5.2). Restoring Y with (3.12) proves (5.3). Optimizers are used only as witnesses for inequalities. □

5.2 One parameter choice verifies every hypothesis

Use the parameters of (1.2), namely

$$q = 2^{20}(n+1), \quad S = nq, \quad d = S+1, \quad H = 2^{20000}d^{40}, \quad W = H^4, \quad B = H^5, \quad o = 514SW.$$

Take

$$\alpha = \beta = \frac{2^{84}d^4}{H}, \qquad a = \alpha + \rho \le \frac{2^{85}d^4}{H}, \qquad b = 104a + \beta. \tag{5.4}$$

Proposition 4.8 gives $s \le o$ and the convenient bounds

$$N_g \le 2^{10}dH^9, \qquad p = Bs \le 2^{80}d^4H^8, \qquad c \le 2^{19}d^2H^8, \qquad h \le 2os \le 2^{91}d^5H^7. \tag{5.5}$$

Also $B \ge 2o$, $B \le o^2/1024$, $o \ge 1000$, $8 \mid B$, and $W/o \le 1/514$. These verify the geometric and core hypotheses. Each follows directly by substituting $W = H^4$ and $B = H^5$ and using $H = 2^{20000}d^{40}$, $d \ge 1$, $S \ge 1$.

In G_0, a grid vertex has at most p outside neighbors: it has no neighbors in C or I. A program fan has at most $p + c$ neighbors outside X, since I is isolated. Thus

$$p/N_g = s/o \le \alpha, \qquad (p+c)/N_g \le \beta, \qquad Q = 12s/o \le \alpha \le a.$$

For the second inequality, $c/N_g < o/(2B)$ and the spare factor in 2^{84} dominates this term together with s/o. The mesh satisfies $\rho \le \alpha$. Finally the scale of H gives $a < 2^{-4096}$ and $b < 1/100$. The two phases, the localized grid estimate, and fan freezing therefore apply to this one actual drawing.

Substitution in their error bounds gives

$$E_g \le 2^{307}d^{14}H^{15}, \qquad E_{\mathrm{fr}} \le 2^{290}d^{14}H^{15}, \qquad R_Y \le 2^{182}d^{10}H^{15}. \tag{5.6}$$

For the freezing term, $Q \le a$, $b \le 105a$, and $b + \rho \le 106a$, whence $E_{\mathrm{fr}} \le 2 \cdot 105 \cdot 106 N_g^2 a^3 < 2^{15}N_g^2a^3$. For R_Y, use $p + c \le 2^{81}d^4H^8$ and (5.5). The resulting product is at most $2^{172}d^9H^{15}$, and $h^2/2 \le 2^{181}d^{10}H^{14}$, both covered by the stated bound. Consequently

$$E_{\mathrm{fr}} + 2E_g + R_Y < 2^{310}d^{14}H^{15}. \tag{5.7}$$

5.3 Quantitative reduction and threshold

Theorem 5.2 (A computable source-value window)**.** *Given a represented permutation graph G with $n \geq 1$, the construction computes a general-position rational point set P, a baseline K_{base}, and a positive integer $\Delta = q^2 H^{16}$, with explicit nonnegative errors such that*

$$K_{\mathrm{base}} + \Delta g(G) - E_- \leq \mathrm{OPT}(P) \leq K_{\mathrm{base}} + \Delta g(G) + E_+, \qquad 2^{18}(E_- + E_+) < \Delta. \tag{5.8}$$

All output data and threshold constants have polynomial binary length and are computable in deterministic polynomial time. The points can instead be returned with integer coordinates.

Proof. Put

$$\begin{aligned} K_{\mathrm{base}} &= K_0 + L_+ + K_C, \\ E_- &= E_{\mathrm{fr}} + E_g, \\ E_+ &= E_g + R_Y + 10BSW^2 + 2SW^4. \end{aligned} \tag{5.9}$$

The core bound (4.12) and Proposition 5.1 give the two-sided interval.

Divide its total error by $\Delta = q^2 H^{16}$. The part in (5.7) is less than

$$\frac{2^{310} d^{14}}{q^2 H} \leq 2^{-19690} d^{-26} < 2^{-40}.$$

The selected unary term satisfies

$$\frac{10BSW^2}{\Delta} = \frac{10n}{qH^3} < 2^{-40},$$

and the conservative local term satisfies

$$\frac{2SW^4}{\Delta} = \frac{2n}{q} < 2^{-19}.$$

Hence the total is less than $2^{-39} + 2^{-19} < 2^{-18}$, proving the required strict window. No sharper local enumeration or narrower aggregate window is used.

To compute the baseline, first enumerate proper crossings on the actual final coordinates by exact determinants. Evaluate c^+, each fan response, the actual fan–fan cost, and each single-flip difference d_v. Evaluate K_C by (4.10): set the palette, use the all-zero source colour as a reference, minimize each short unary separately, insert the known dummy value, subtract the explicit $\Delta|E(G)|$, and add the residual unary minima. Finally insert $Z(B)$. These are finite sums and minima of two known integers. The source optimum, core optimum, and target optimum never occur as computational subroutines.

The output point count is $N = B + o + s \leq 3H^5$. Since $S = 2^{20} n(n+1)$, it is bounded by a fixed polynomial of degree 400 in n. Proposition 3.8 proves polynomial coordinate bit length and construction time; it also gives the integer-coordinate version by a common positive scaling. □

We make the bit bound for the decision threshold explicit as well. For actual total point count N, all grid, fan, and core edge counts are at most N^2. Each unnormalized crossing contribution is zero or one, and all fields are differences of finite such sums. The baseline formula (5.1) and (4.10) therefore gives the deliberately loose bound

$$|K_{\mathrm{base}}| \leq 16N^6.$$

One may also see this by computing each baseline from its fixed colouring cost and at most N^2 single-edge unary corrections, each bounded by a fixed multiple of N^2; $\Delta|E(G)|$ is the all-zero main crossing count, at most N^4. Since $q, H \le N$, we have $\Delta \le N^{18}$, and $|E(G)| \le N^2$.

For a source MaxCut threshold $k \in \mathbb{Z}$, put $t = |E(G)| - k$ and output

$$\kappa = \lfloor K_{\text{base}} + \Delta t + \Delta/4 \rfloor. \tag{5.10}$$

In particular

$$|\kappa| \le 32N^{20}(|k| + 1), \tag{5.11}$$

so the threshold bit length is $O(\log N + \text{bits}(k))$. The baseline is integral in the counting presentation above; the floor also accommodates its equivalent rational presentation in the formalization.

Proof of Theorem 1.2. Use the NP-hard represented permutation-graph problem with threshold k, or equivalently $g(G) \le t$. For $n \ge 1$ apply Theorem 5.2 and output (5.10). If $g(G) \le t$, then

$$\text{OPT}(P) \le K_{\text{base}} + \Delta t + E_+ < K_{\text{base}} + \Delta t + \Delta/4.$$

Since the objective is an integer, it is at most κ. If $g(G) > t$, integrality gives $g(G) \ge t + 1$, and

$$\text{OPT}(P) \ge K_{\text{base}} + \Delta(t + 1) - E_- > K_{\text{base}} + \Delta t + \Delta/4 \ge \kappa.$$

Thus the same point set and threshold give both directions.

Empty source inputs can be decided directly. A fixed yes instance consists of two distinct integer points and threshold zero. A fixed no instance is the five points (i, i^2), $0 \le i < 5$, with threshold zero: their diagonal crossing graph is a five-cycle and cannot be properly two-coloured. If the target convention permits only nonnegative thresholds and the computed κ is negative, output this fixed no instance. A source yes instance would otherwise imply the impossible inequality $0 \le \text{OPT}(P) \le \kappa < 0$. Thresholds outside the source edge-count range may also be handled directly, though the displayed equivalence already covers them.

Proposition 3.8 and (5.11) prove a deterministic polynomial-time many-one reduction with a single complete geometric target instance. Every target edge colour is a free certificate bit. The palette and frozen colours enter the proof through minimizer properties and one compatible upper-bound colouring, not through precolouring restrictions on the output. Scaling produces the same hardness on integer coordinates. Membership in NP follows by assigning colours to all $\binom{N}{2}$ edges and testing the $O(N^4)$ edge pairs with exact orientation determinants. This proves NP-completeness and, consequently, NP-hardness of exact minimization. □

Corollary 5.3. *If collinear input points are permitted while the objective still counts only proper crossings using strict orientation tests, the decision problem remains NP-complete.*

Proof. The general-position hard instances are also inputs in the larger class. The same polynomial verifier excludes every zero orientation determinant and therefore handles the stipulated crossing convention on degenerate inputs. No perturbation of an arbitrary degenerate input is used. □

6 Formal verification and reproducible computations

The accompanying Lean development formalizes the new mathematical construction and its final interface with the complete geometric graph problem. The principal declaration is

```
Q4.Candidate.ConstructedReduction.
    constructed_maxcut_instance.
```

For every $n \geq 1$, permutation π and integer source threshold k, it produces one point count N, one coordinate function $\mathrm{coords} : \mathrm{Fin}(N) \to \mathbb{Q}^2$, and one integer threshold κ. The same witness satisfies injectivity, general position, coordinate encoding bounds and the equivalence

$$\exists \chi\ M(\mathrm{coords}, \chi) \leq \kappa \quad \Longleftrightarrow \quad \exists \text{ a cut of the inversion graph of } \pi \text{ of size at least } k.$$

The colours on the left range over every edge of K_N freely. The formal construction gives, with $\widehat{H} = 2^{20840}(n+1)^{80}$,

$$N \leq 3\widehat{H}^5, \qquad L_{\mathrm{coord}} \leq 2^{17}\widehat{H}^{10},$$

where each coordinate has an integer numerator and nonzero denominator of absolute value at most $2^{L_{\mathrm{coord}}}$.

6.1 What the theorem assumes and what it proves

The final Lean declaration has the explicit mathematical hypothesis

```
forall m, ConvexOptimum m (knownZ m).
```

Here `ConvexOptimum` asserts both a lower bound for every free two-colouring of the chords and existence of an attaining colouring; `knownZ` is the value Z above. This hypothesis is supplied in the mathematical proof by the established two-page theorem [1]. Its conversion to the actual frame and dummy chord objectives is formalized. The literature theorem itself is not reproved in this development and is not introduced as a custom Lean axiom.

All other construction obligations in the final declaration are instantiated: actual program multiplicities, geometry, general-position placement, palette forcing in the appropriate core optimization, selected-edge residual fields, source amplification, the complete edge partition, the computable background, and the strict error gap. In particular, an unknown source optimum or background optimum is not an input to the reduction.

Proof component	Principal module families
Analytic and discrete stability	`TwoPhaseStability`, `GridStepBridge`, `GridLocalization`
Complete energy and freezing	`CompleteProbeFreezing`, `CompleteClassEnergy`, `CompleteBackground`
Actual geometric programs	`ProgramGeometry`, `ActualProgramGeometry`, `ActualRecipe`, `ProgramPlacement`
Free core and source signal	`ActualCatalogBounds`, `ActualSelectedUnary`, `MainSourceEncoding`, `CoreValueBridge`
Effective baseline and encoding	`ComputedBackground`, `ComputedCore`, `RationalEncoding`, `CandidateSize`
Final complete-input equivalence	`CompleteInput`, `ActualReductionValue`, `SourceCutBridge`, `ConstructedReduction`

The development uses the conservative local bound $2SW^4$ and proves $2^{18}(E_- + E_+) < \Delta$. These are the bounds used in the final reduction here. No stronger constant from an earlier draft is needed.

The source NP-hardness theorem [2], complexity classes, machine running times, membership in NP, and the separate output-threshold bit bound are not formalized in Lean. Their mathematical proofs and references are given in this paper and in the effective-reduction notes. In particular, a small-coordinate existence theorem alone would not prove the required polynomial construction time; the explicit finite selection algorithm supplies this additional fact.

6.2 Verification procedure

The public proof artifact contains exactly the transitive project import closure of the final theorem: 123 construction modules and four shared foundation modules, with a separate explicit aggregate entry point. The Lean version is fixed to 4.34.1 and the mathlib dependency and its transitive revisions are fixed by the Lake manifest.

The verification script first compares the source set with this closure and its SHA256 manifest. It checks for proof placeholders and custom axioms, builds the aggregate entry point, elaborates every source file individually, and checks the outputs of the retained explicit `#print axioms` commands, including the final theorem. It also checks that source and configuration hashes did not change during the run. The permitted kernel axiom dependencies are `propext`, `Classical.choice`, and `Quot.sound`. The axiom queries cover the designated key declarations; they are not described as a separate enumeration of every declaration in mathlib. The trust boundary includes Lean, the pinned dependencies and their compiled artifacts, and the ordinary compiler and hardware stack.

6.3 Finite diagnostics and review scope

Four standard-library Python programs provide exact finite diagnostics: the six-point completion obstruction in Appendix A; core combinatorics for all six orders of three source vertices; two reduced geometric realizations with actual general-position insertion and crossing-table checks; and rational or integer checks of selected parameter inequalities. The programs use exact arithmetic. Their recorded inputs and expected outputs are distributed with the source. They do not generate or solve the extremely large target instances of the reduction and are not used as proofs of statements for unbounded input sizes.

AI systems assisted the exploration, formalization and preparation of the manuscript. Independent LLM-assisted reviews were used to examine definitions, interfaces and the effective construction, followed by kernel checks of the formal development. The artifact records this scope explicitly; these reviews do not constitute human journal peer review. The repository retains the final proof and useful diagnostics, with intermediate exploratory records kept outside the public source set.

A A direct-completion obstruction

The following example explains why the contribution of added edges must be controlled in a completeness-preserving reduction.

Proposition A.1. *Completing a segment family to the complete geometric graph on its endpoints can produce an optimum whose restriction to the original family is not optimal. The minimum contribution of the added edges can also depend on the prescribed source colours.*

Proof. Take $p_i = (i, i^2)$ for $0 \le i < 6$ and abbreviate $p_i p_j$ by ij. The source edges are $01, 24, 35$. Their only crossing pair is $24, 35$, so the source optimum is zero.

Every five-point subset is strictly convex. The crossing graph of its five diagonals is an odd cycle, so every two-colouring contains a monochromatic crossing. There are six five-point subsets, and every crossing pair belongs to exactly two of them. Therefore the complete six-point graph has at least three monochromatic crossing pairs in every colouring.

Colour $03, 04, 13, 14$ blue and all remaining edges red. Its monochromatic crossing pairs are exactly $(02, 15), (03, 14), (24, 35)$, so it is optimal for the complete graph. Its restriction to the source has cost one. In particular, when 24 and 35 have the same colour, the minimum additional cost is two.

Alternatively colour $14, 15, 24, 25$ blue and the other edges red. The monochromatic crossing pairs are $(02, 13), (04, 35), (14, 25)$. This is again a complete optimum, but 24 and 35 now have different colours. The source cost is zero and the minimum additional cost is three. Hence the extra optimum is not independent of the source colouring. □

References

[1] Bernardo M. Ábrego, Oswin Aichholzer, Silvia Fernández-Merchant, Pedro Ramos, and Gelasio Salazar. The 2-page crossing number of K_n. *Discrete & Computational Geometry*, 49(4):747–777, 2013. Open version: arXiv:1206.5669.

[2] Celina M. H. de Figueiredo, Alexsander A. de Melo, Fabiano S. Oliveira, and Ana Silva. MaxCut on permutation graphs is NP-complete. *Journal of Graph Theory*, 104(1):5–16, 2023. Open version: arXiv:2202.13955.

[3] Robert Ganian, Fabrizio Montecchiani, Martin Nöllenburg, and Meirav Zehavi. Parameterized complexity in graph drawing (Dagstuhl seminar 21293). *Dagstuhl Reports*, 11(6):82–123, 2021. Editors; see Vogtenhuber's Problem 2 in Section 4.12, p. 107.